

Contents

Chapter 1

Mott Criticality as the Confinement Transition of a Pseudogap-Mott Metal: Insights on Mott Transition and Pseudogap in Two Dimensions

In the last chapter, we developed a new auxiliary model framework for mapping the low-energy physics of quantum impurity models to that of lattice models. This led to relations between correlation functions, Greens functions and entanglement measures, between the two models. We devote this chapter to applying this method to the problem of Mott transition on the 2D square lattice at half-filling. We first motivate that the appropriate auxiliary model is an extended Anderson impurity model (of the kind studied in Chapter 3) but embedded into a square lattice. This allows encoding the lattice geometry into the impurity model couplings. Through the tiling framework, we reveal that the passage from the Fermi liquid into the Mott insulator involves a distinct intervening pseudogapped phase of matter that has anomalous non-Fermi liquid properties and holon-doublon excitations. This provides a compelling picture of the Mott transition as the confinement of pairs of holes and doubles.

1.1 Introduction

The Landau quasiparticle paradigm [1] underpins our understanding of metallic behaviour, where the screening of long-ranged repulsive interactions between delocalised electrons leads to well-defined excitations that carry the same quantum numbers as free electrons. This remarkable phenomenon - known as adiabatic continuity - is captured by excitations described by poles of the single-particle Greens function of the interacting system. But what defines metallicity when quasiparticles are no longer well-defined [2]? Can a surface of zeros of the Greens function [3], rather than poles forming a Fermi surface, define an entirely new kind of metal? Could such an exotic state serve as the precursor to Mott localisation driven purely by repulsion, and without invoking magnetic order [4]? These questions necessitate a radical departure in understanding conduction in

correlated electron systems beyond the quasiparticle paradigm [5].

The anomalous pseudogap (PG) phase, observed in the cuprates, exhibits nodal-antinodal dichotomy with spectral weight concentrated on Fermi arcs around the nodal regions [6–8]. In the PG phase of such Mott systems [9–38], the single-particle Green’s function develops surfaces of zeros, also known as Luttinger surfaces [3, 39], that signals a fundamental obstruction to quasiparticle formation. As interaction strength increases, these zero surfaces proliferate, fragmenting the Fermi surface into nodal arcs [18, 39], violating the Luttinger sum rule by reconfiguring Fermi volume [40], and breaking an emergent $\mathbb{Z}_2$ symmetry associated with the conservation of spin currents in Fermi liquids [41, 42]. Unlike conventional instabilities driven by poles, this phenomenon marks a topological breakdown of Fermi liquid theory [2, 43]. We show that this breakdown arises from a strongly entangled, scale-invariant state of quantum matter, one that serves as the parent metal to the Mott insulator in a half-filled lattice model of strongly correlated electrons. This offers a concrete, controlled framework to understand both the non-quasiparticle character of the pseudogap phase and its continuous evolution into a repulsion-driven insulator, in full alignment with Mott’s original vision [4].

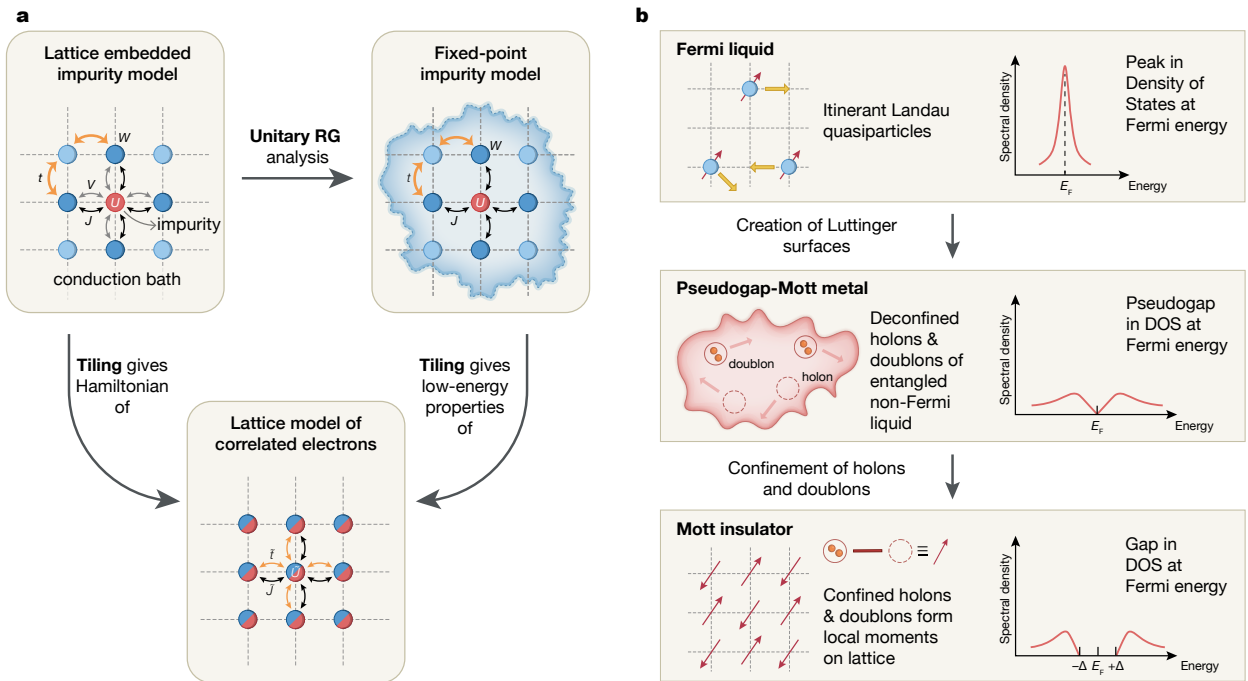


Figure 1.1: Schematic representation of (a) theoretical framework adopted by us (see main text for explanation of symbols), (b) phases obtained for the extended Hubbard model in two dimensions at $T = 0$, and passage between them.

Our approach involves developing a renormalization group (RG)-based impurity-to-lattice framework that reconstructs the low-energy physics of a strongly correlated system from the fixed-point structure of a quantum impurity model (Fig. 1.1(a)). Starting from a dynamical Anderson impurity

embedded in a correlated bath, we solve the system using a unitary RG (URG) approach [44,45]. We then construct a translation-invariant lattice Hamiltonian through a many-body tiling procedure that preserves non-local correlations and inherits momentum-resolved self-energies and renormalized couplings from the impurity solution. The resulting model is an *extended*-Hubbard Hamiltonian on a two-dimensional square lattice at half-filling, with both on-site repulsion and nearest-neighbour spin exchange interactions. These correlations, derived directly from the impurity dynamics, enable tracking the evolution from the FL to the Mott insulator (MI) via an intermediate PG phase.

A central insight of our work is that the PG phase hosts a NFL state governed by Kondo frustration and doublon–holon deconfinement [46] (Fig. 1.1(b)). A correlation scale (W) in the conduction bath suppresses Kondo screening (with coupling J) beyond a critical ratio of W/J , triggering the emergence of Luttinger surfaces in the lattice model. This breakdown of quasiparticles begins at the antinodes and progressively engulfs the nodal regions, driving a continuous transformation into the MI. The resulting NFL exhibits a pseudogapped density of states that vanishes as ω^2 as $\omega \rightarrow 0$, multipartite long-range entanglement, and a universal scattering rate of the form $\sim (a + b\omega^2)^{-1}$ that grows as $\omega \rightarrow 0$. At the critical endpoint of this phase, the URG flow identifies a singular NFL described by the Hatsugai–Kohmoto (HK) model [42,47,48], with fully deconfined holon–doublon excitations.

The emergence of Luttinger surfaces alters the anomaly structure associated with the generalized symmetry of the FS [49–51]. The corresponding topological charge is defined via the Luttinger–Ward functional. This charge is modified by Green’s function zeros, signalling a shift in the underlying anomaly. Within our framework, this reorganization grants topological protection to the RG flows terminating in the PG regime, ensuring the stability of its NFL excitations. These are adiabatically connected to the Hatsugai–Kohmoto model [47,48], reinforcing the interpretation of the PG as a distinct quantum phase. We thus identify the NFL state as a new gapless, long-range entangled phase of correlated matter - the *Mott metal* - whose deconfined holon-doublon excitations are eventually confined across the Mott transition [4].

While we demonstrate our approach for an extended Hubbard model, the underlying framework is broadly applicable to correlated quantum systems. The impurity-plus-tiling construction accommodates multi-orbital degrees of freedom, frustrated geometries, and nontrivial band topology. By deriving lattice Hamiltonians directly from renormalized local dynamics - without relying on mean-field approximations - this method preserves nonlocal correlations and enhances momentum-space resolution. Coupled with the unitary renormalisation group method as the impurity solver, this approach provides access to one-particle as well as two-particle correlations, entanglement measures and several dynamical quantities, making the method very versatile. It thus provides a versatile and controlled route for engineering strongly correlated models, particularly in regimes where PG formation, NFL scaling, and Mott physics intersect.

We now describe the layout of the work. In section 1.2, we formulate the lattice-embedded auxiliary quantum impurity model and sketch the unitary renormalisation group (URG) method employed to analyse the impurity model. In Chapter ??, we have already formulated the tiling procedure by which to map the impurity Hamiltonian, eigenstates and observables into a lattice model of strongly correlated electrons via the usage of many-body translation operators. Section 1.5 displays the destruction of the Fermi liquid from the breakdown of Kondo screening, leading to a pseudogapped non-Fermi liquid phase of matter. The novel properties of this phase are rigorously

established in section 1.6. In section 1.7, we demonstrate that the continuous transition between the pseudogap and Mott insulating phases is described by the exactly solvable Hatsugai-Kohmoto model. We demonstrate in section 1.9 the universal organising principle that is responsible for the emergence of the pseudogap non-Fermi liquid phase. This also unveils the pseudogap as the strongly coupled long-range entangled parent metal of the Mott insulator. We conclude in section 1.10. Finally, we present appendices containing the derivation of the URG analysis of the impurity model, and the exactly solvable model at the critical point.

1.2 Auxiliary Model and URG analysis

To capture the interplay between Kondo physics and Mott localization, our approach relies on a two-dimensional auxiliary model consisting of an Anderson impurity embedded in a minimally-correlated bath, which we solve the system using a unitary RG (URG) approach [44, 45]. The impurity model and URG analysis are laid out in this section. In order to make contact with a lattice model, we construct a translation-invariant Hamiltonian from the impurity model through a many-body tiling procedure: this preserves non-local correlations, as well as allows for momentum-resolved self-energies and renormalized couplings generated from the impurity solution. For the impurity model chosen in this work, tiling results in an *extended*-Hubbard Hamiltonian on a two-dimensional square lattice at particle-hole symmetry (half-filling). The interaction terms within the lattice model are a result of analogous correlations present in the auxiliary model. This is laid out next in Section ??.

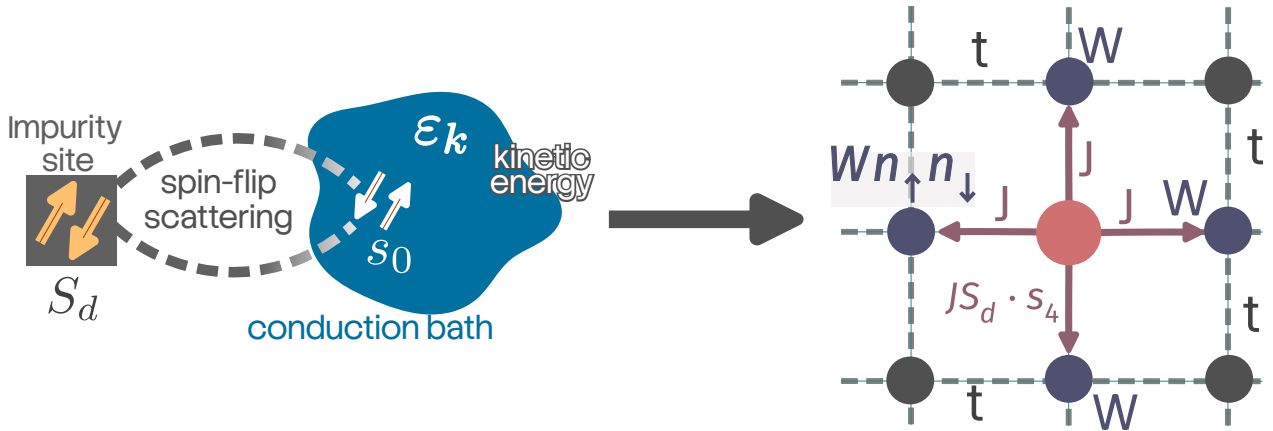


Figure 1.2: Particle and hole states.

1.2.1 Lattice-embedded Impurity Model

The auxiliary model consists of a correlated impurity site (with a double occupancy cost U) embedded within a correlated conduction bath defined on a square lattice (see Fig. 1.1(a)). The impurity site hybridises with the conduction bath sites adjacent to it through one-particle hybridisation V as

well as a spin-exchange interaction J arising from fluctuations in local spin densities of the electrons. The conduction bath has minimal correlations in the form of a local correlation term W on the sites that are adjacent to the impurity.

The Hamiltonian of the auxiliary model studied here is

$$\mathcal{H}_{\text{aux}} = H_{\text{imp}} + H_{\text{coup}} + H_{\text{cbath}}, \quad (2.1)$$

where the particle-hole symmetric impurity term is

$$H_{\text{imp}} = -\frac{U}{2} (\hat{n}_{d\uparrow} - \hat{n}_{d\downarrow})^2, \quad (2.2)$$

describing on-site Coulomb interactions at the impurity. Here, $c_{d\sigma}^\dagger$ creates an electron with spin σ at the impurity site, and $\hat{n}_{d\sigma} = c_{d\sigma}^\dagger c_{d\sigma}$ denotes the corresponding number operator.

The impurity couples to four nearest-neighbour bath sites $c_{Z\sigma}$ through both hybridization and Kondo exchange:

$$H_{\text{coup}} = \frac{J}{2} \sum_{\sigma_1, \sigma_2, Z} \mathbf{S}_d \cdot c_{Z\sigma_1}^\dagger \boldsymbol{\tau}_{\sigma_1 \sigma_2} c_{Z\sigma_2} - V \sum_{\sigma, Z} \left(c_{Z\sigma}^\dagger c_{d\sigma} + \text{h.c.} \right), \quad (2.3)$$

where $\boldsymbol{\tau}$ are the Pauli matrices, and Z indexes the four neighbouring bath sites. The half-filled bath includes nearest neighbour hopping and on-site correlations on the site neighbouring the impurity:

$$H_{\text{cbath}} = \sum_{\mathbf{k}} \epsilon_{\mathbf{k}} c_{\mathbf{k},\sigma}^\dagger c_{\mathbf{k},\sigma} - \frac{W}{2} \sum_Z (\hat{n}_{Z\uparrow} - \hat{n}_{Z\downarrow})^2, \quad (2.4)$$

where $\epsilon_{\mathbf{k}} = -2t(\cos k_x + \cos k_y)$, and W parametrizes local repulsion in the bath that frustrates Kondo screening and drives the system towards local-moment formation. A crucial feature of this construction is that the Kondo coupling acquires a momentum structure upon Fourier transforming:

$$J_{\mathbf{k},\mathbf{k}'} = \frac{J}{2} [\cos(k_x - k'_x) + \cos(k_y - k'_y)], \quad (2.5)$$

which respects the C_4 symmetry of the square lattice.

1.2.2 The unitary renormalisation group method

In order to obtain the various low-energy phases of our impurity model, we perform a scaling analysis of the associated Hamiltonian using the recently developed unitary renormalisation group (URG) method [19,39,40]. The method has been applied successfully on a wide variety of problems of correlated fermions [?, 52–56]. The method proceeds by resolving quantum fluctuations in high-energy degrees of freedom, leading to a low-energy Hamiltonian with renormalised couplings and new emergent degrees of freedom. Typically, for a system with Fermi energy ϵ_F and bandwidth E_N , the sequence of isoenergetic shells $\{E_{(j)}\}$, $E_j \in [E_0, E_N]$ define the states whose quantum fluctuations we sequentially resolve. The momentum states lying on shells E_N that are far away from the

Fermi surface comprise the UV states, while those on shells near the Fermi surface comprise the IR states.

As a result of the URG transformations, the Hamiltonian $H_{(j)}$ at a given RG step j involves scattering processes between the k -states that have energies lower than $D_{(j+1)}$. The unitary transformation $U_{(j)}$ is then defined so as to remove the number fluctuations of the currently most energetic set of states $D_{(j)}$ [44,45]:

$$H_{(j-1)} = U_{(j)} H_{(j)} U_{(j)}^\dagger, \text{ such that } [H_{(j-1)}, \hat{n}_j] = 0. \quad (2.6)$$

The eigenvalue of $\hat{n}_j$ has, thus, been rendered an integral of motion (IOM) under the RG transformation.

The unitary transformations can be expressed in terms of a generator $\eta_{(j)}$ that has fermionic algebra [44,45]:

$$U_{(j)} = \frac{1}{\sqrt{2}} \left(1 + \eta_{(j)} - \eta_{(j)}^\dagger \right), \quad \{ \eta_{(j)}, \eta_{(j)}^\dagger \} = 1, \quad (2.7)$$

where $\{ \cdot \}$ is the anticommutator. The unitary operator $U_{(j)}$ that appears in Eq. (2.7) can be cast into the well-known general form $U = e^{\mathcal{S}}$, $\mathcal{S} = \frac{\pi}{4} \left(\eta_{(j)}^\dagger - \eta_{(j)} \right)$ that a unitary operator can take, defined by an anti-Hermitian operator $\mathcal{S}$. The generator $\eta_{(j)}$ is given by the expression [44,45]

$$\eta_{(j)}^\dagger = \frac{1}{\hat{\omega}_{(j)} - \text{Tr}(H_{(j)} \hat{n}_j)} c_j^\dagger \text{Tr}(H_{(j)} c_j). \quad (2.8)$$

The operators $\eta_{(j)}, \eta_{(j)}^\dagger$ behave as the many-particle analogues of the single-particle field operators $c_j, c_j^\dagger$ - they change the occupation number of the single-particle Fock space $|n_j\rangle$. The important operator $\hat{\omega}_{(j)}$ originates from the quantum fluctuations that exist in the problem because of the non-commutation of the kinetic energy terms and the interaction terms in the Hamiltonian:

$$\hat{\omega}_{(j)} = H_{(j-1)} - H_{(j)}^i. \quad (2.9)$$

$H_{(j)}^i$ is the part of $H_{(j)}$ that commutes with $\hat{n}_j$ but does *not* commute with at least one $\hat{n}_l$ for $l < j$. The RG flow continues up to energy D^* , where a fixed point is reached from the vanishing of the RG function. Detailed comparisons of the URG with other methods (e.g., the functional RG, spectrum bifurcation RG etc.) can be found in Refs. [44,53], and implementation on the present problem in Appendix A.

1.3 Derivation of unitary RG equations for the lattice-embedded impurity model

1.3.1 RG scheme

At any given step j of the RG procedure, we decouple the states $\{\mathbf{q}\}$ on the isoenergetic surface of energy ε_j . The diagonal Hamiltonian H_D for this step consists of all terms that do not change the

occupancy of the states $\{\mathbf{q}\}$:

$$H_D^{(j)} = \varepsilon_j \sum_{q,\sigma} \tau_{q,\sigma} + \frac{1}{2} \sum_{\mathbf{q}} J_{\mathbf{q},\mathbf{q}} S_d^z (\hat{n}_{\mathbf{q},\uparrow} - \hat{n}_{\mathbf{q},\downarrow}) - \frac{1}{2} \sum_{\mathbf{q}} W_{\mathbf{q}} (\hat{n}_{\mathbf{q},\uparrow} - \hat{n}_{\mathbf{q},\downarrow})^2, \quad (3.1)$$

where $\tau = \hat{n} - 1/2$ and $W_{\mathbf{q}}$ is a shorthand for $W_{\mathbf{q},\mathbf{q},\mathbf{q},\mathbf{q}}$. The three terms, respectively, are the kinetic energy of the momentum states on the isoenergetic shell that we are decoupling, the spin-correlation energy between the impurity spin and the spins formed by these momentum states and, finally, the local correlation energy associated with these states arising from the W term. The off-diagonal part of the Hamiltonian on the other hand leads to scattering in the states $\{\mathbf{q}\}$. We now list these terms, classified by the coupling they originate from.

Arising from the Kondo spin-exchange term

$$\begin{aligned} T_{KZ1}^\dagger + T_{KZ1} &= \frac{1}{2} \sum_{\mathbf{k},\mathbf{q},\sigma} \sigma J_{\mathbf{k},\mathbf{q}} S_d^z \left[c_{\mathbf{q}\sigma}^\dagger c_{\mathbf{k},\sigma} + \text{h.c.} \right], \\ T_{KZ2}^\dagger + T_{KZ2} &= \frac{1}{2} \sum_{\mathbf{q},\sigma} \sigma J_{\mathbf{q},\bar{\mathbf{q}}} S_d^z \left[c_{\mathbf{q}\sigma}^\dagger c_{\bar{\mathbf{q}},\sigma} + \text{h.c.} \right], \\ T_{KT1}^\dagger + T_{KT1} &= \frac{1}{2} \sum_{\mathbf{k},\mathbf{q}} J_{\mathbf{k},\mathbf{q}} \left[S_d^+ \left(c_{\mathbf{q}\downarrow}^\dagger c_{\mathbf{k}\uparrow} + c_{\mathbf{k}\downarrow}^\dagger c_{\mathbf{q}\uparrow} \right) + \text{h.c.} \right], \\ T_{KT2}^\dagger + T_{KT2} &= \frac{1}{2} \sum_{\mathbf{q}} J_{\mathbf{q},\bar{\mathbf{q}}} \left[S_d^+ \left(c_{\mathbf{q}\downarrow}^\dagger c_{\bar{\mathbf{q}}\uparrow} + c_{\bar{\mathbf{q}}\downarrow}^\dagger c_{\mathbf{q}\uparrow} \right) + \text{h.c.} \right], \end{aligned} \quad (3.2)$$

Arising from spin-preserving scattering within conduction bath

$$\begin{aligned} T_{P1}^\dagger + T_{P1} &= - \sum_{\mathbf{q} \in \varepsilon_j} \sum_{\mathbf{k}_2, \mathbf{k}_3, \mathbf{k}_4 < \varepsilon_j} \sum_{\sigma} \left[W_{\mathbf{q}, \mathbf{k}_2, \mathbf{k}_3, \mathbf{k}_4} c_{\mathbf{q},\sigma}^\dagger c_{\mathbf{k}_2,\sigma} c_{\mathbf{k}_3,\sigma}^\dagger c_{\mathbf{k}_4,\sigma} + \text{h.c.} \right] \\ T_{P2}^\dagger + T_{P3} &= - \sum_{\mathbf{q} \in \varepsilon_j} \sum_{\mathbf{k}_2 < \varepsilon_j} \sum_{\sigma} W_{\mathbf{q}, \mathbf{k}_2, \bar{\mathbf{q}}, \bar{\mathbf{q}}} c_{\mathbf{q},\sigma}^\dagger c_{\mathbf{k}_2,\sigma} n_{\bar{\mathbf{q}},\sigma} - \sum_{\mathbf{q} \in \varepsilon_j} \sum_{\mathbf{k}_1 < \varepsilon_j} \sum_{\sigma} W_{\mathbf{k}_1, \mathbf{q}, \mathbf{q}, \mathbf{q}} c_{\mathbf{k}_1,\sigma}^\dagger c_{\mathbf{q},\sigma} n_{\mathbf{q},\sigma} \\ T_{P4} &= - \sum_{\mathbf{q} \in \varepsilon_j} \sum_{\mathbf{k}_2, \mathbf{k}_3 < \varepsilon_j} \sum_{\sigma} W_{\mathbf{q}, \bar{\mathbf{q}}, \mathbf{k}_2, \mathbf{k}_3} c_{\mathbf{q},\sigma}^\dagger c_{\bar{\mathbf{q}},\sigma} c_{\mathbf{k}_2,\sigma}^\dagger c_{\mathbf{k}_3,\sigma} \\ T_{P5} &= - \sum_{\mathbf{q} \in \varepsilon_j} \sum_{\mathbf{k}_2, \mathbf{k}_3 < \varepsilon_j} \sum_{\sigma} W_{\mathbf{q}, \mathbf{k}_2, \mathbf{k}_3, \bar{\mathbf{q}}} c_{\mathbf{q},\sigma}^\dagger c_{\mathbf{k}_2,\sigma} c_{\mathbf{k}_3,\sigma}^\dagger c_{\bar{\mathbf{q}},\sigma} \\ &= + \sum_{\mathbf{q} \in \varepsilon_j} \sum_{\mathbf{k}_2, \mathbf{k}_3 < \varepsilon_j} \sum_{\sigma} W_{\mathbf{q}, \mathbf{k}_3, \mathbf{k}_2, \bar{\mathbf{q}}} c_{\mathbf{q},\sigma}^\dagger c_{\bar{\mathbf{q}},\sigma} c_{\mathbf{k}_2,\sigma}^\dagger c_{\mathbf{k}_3,\sigma} \\ &= -T_{P4} \end{aligned} \quad (3.3)$$

Arising from spin-flip scattering within conduction bath

$$\begin{aligned}
 T_{F1}^\dagger + T_{F1} &= \sum_{\mathbf{q} \in \varepsilon_j} \sum_{\mathbf{k}_2, \mathbf{k}_3, \mathbf{k}_4 < \varepsilon_j} \sum_{\sigma} \left[W_{\mathbf{q}, \mathbf{k}_2, \mathbf{k}_3, \mathbf{k}_4} c_{\mathbf{q}, \sigma}^\dagger c_{\mathbf{k}_2, \sigma} c_{\mathbf{k}_3, \bar{\sigma}}^\dagger c_{\mathbf{k}_4, \bar{\sigma}} + \text{h.c.} \right] \\
 T_{F2} &= \sum_{\mathbf{q}, \mathbf{q}' \in \varepsilon_j} \sum_{\mathbf{k}_2, \mathbf{k}_3 < \varepsilon_j} \sum_{\sigma} W_{\mathbf{q}, \mathbf{q}', \mathbf{k}_2, \mathbf{k}_3} c_{\mathbf{q}, \sigma}^\dagger c_{\mathbf{q}', \sigma} c_{\mathbf{k}_2, \bar{\sigma}}^\dagger c_{\mathbf{k}_3, \bar{\sigma}} \\
 T_{F3} &= \sum_{\mathbf{q}, \mathbf{q}' \in \varepsilon_j} \sum_{\mathbf{k}_2, \mathbf{k}_3 < \varepsilon_j} \sum_{\sigma} W_{\mathbf{q}, \mathbf{k}_2, \mathbf{k}_3, \mathbf{q}'} c_{\mathbf{q}, \sigma}^\dagger c_{\mathbf{k}_2, \sigma} c_{\mathbf{k}_3, \bar{\sigma}}^\dagger c_{\mathbf{q}', \bar{\sigma}} \\
 T_{F4}^\dagger + T_{F4} &= \sum_{\mathbf{q}, \mathbf{q}' \in \varepsilon_j} \sum_{\mathbf{k}_1 < \varepsilon_j} \sum_{\sigma} \left[W_{\mathbf{q}, \mathbf{q}', \mathbf{k}_1} n_{\mathbf{q}, \sigma} c_{\mathbf{q}', \bar{\sigma}}^\dagger c_{\mathbf{k}_1, \bar{\sigma}} + \text{h.c.} \right]
 \end{aligned} \tag{3.4}$$

In all of the terms $T_{P[i]}$ and $T_{F[i]}$, the factor of $1/2$ in front has been cancelled out by a factor of 2 coming from the multiple possibilities of arranging the momentum labels. We will henceforth ignore T_{P4} and T_{P5} because they cancel each other out.

The renormalisation of the Hamiltonian is constructed from the general expression

$$\Delta H^{(j)} = H_X \frac{1}{\omega - H_D} H_X . \tag{3.5}$$

The states on the isoenergetic shell $\pm|\varepsilon_j|$ come in particle-hole pairs $(\mathbf{q}, \bar{\mathbf{q}})$ with energies of opposite signs (relative to the Fermi energy). If $\mathbf{q}$ is defined as the hole state (unoccupied in the absence of quantum fluctuations), it will have positive energy, while the particle state $\bar{\mathbf{q}}$ will be of negative energy and hence below the Fermi surface. To be more specific, given a state $\mathbf{q}$ with energy $\pm|\varepsilon_j|$, we define its particle-hole transformed counterpart as the state $\bar{\mathbf{q}} = \pi + \mathbf{q}$, having energy $\mp|\varepsilon_j|$ and residing in the opposite quadrant of the Brillouin zone. Given this definition, we have the important property that

$$\begin{aligned}
 J_{\mathbf{k}, \bar{\mathbf{q}}} &= -J_{\mathbf{k}, \mathbf{q}}, \\
 W_{\{\mathbf{k}\}, \bar{\mathbf{q}}} &= -W_{\{\mathbf{k}\}, \mathbf{q}} .
 \end{aligned} \tag{3.6}$$

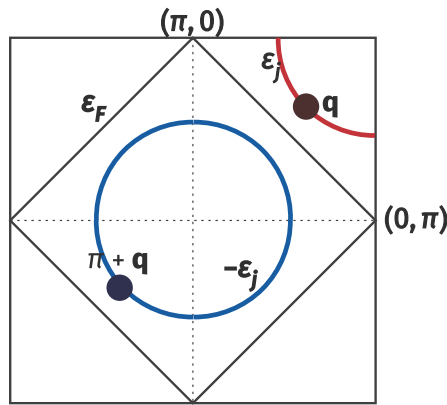


Figure 1.3: Particle and hole states.

1.3.2 Renormalisation of the bath correlation term W

The bath correlation term W can undergo renormalisation only via scattering processes arising from itself. Irrespective of whether the state $\mathbf{q}$ being decoupled is in a particle or hole configuration in the initial many-body state, the propagator $G = 1/(\omega - H_D)$ of the intermediate excited state is uniform, and equal to

$$G_W = 1/(\omega - |\varepsilon_j|/2 + W_{\mathbf{q}}/2) , \quad (3.7)$$

where $W_{\mathbf{q}}$ is the same whether $\mathbf{q}$ is above or below the Fermi surface. The $|\varepsilon_j|/2$ in H_D arises from the excited nature of the state after the initial scattering process.

Scattering arising purely from spin-preserving processes

In this subsection, we calculate the renormalisation to W arising from the terms T_{P1} , T_{P2} and T_{P3} . The first term is

$$\begin{aligned} T_{P1}^\dagger G_W T_{P3} &= \sum_{\sigma} \sum_{\mathbf{k}_1, \mathbf{k}_2, \mathbf{k}_3, \mathbf{k}_4} \sum_{\mathbf{q}} W_{\mathbf{q}, \mathbf{k}_2, \mathbf{k}_3, \mathbf{k}_4} c_{\mathbf{q}, \sigma}^\dagger c_{\mathbf{k}_2, \sigma} c_{\mathbf{k}_3, \sigma}^\dagger c_{\mathbf{k}_4, \sigma} G_W W_{\mathbf{k}_1, \mathbf{q}, \mathbf{q}, \mathbf{q}} c_{\mathbf{k}_1, \sigma}^\dagger c_{\mathbf{q}, \sigma} n_{\mathbf{q}, \sigma} \\ &= - \sum_{\sigma} \sum_{\mathbf{k}_1, \mathbf{k}_2, \mathbf{k}_3, \mathbf{k}_4} c_{\mathbf{k}_1, \sigma}^\dagger c_{\mathbf{k}_2, \sigma} c_{\mathbf{k}_3, \sigma}^\dagger c_{\mathbf{k}_4, \sigma} \sum_{\mathbf{q} \in \text{PS}} W_{\mathbf{q}, \mathbf{k}_2, \mathbf{k}_3, \mathbf{k}_4} G_W W_{\mathbf{k}_1, \mathbf{q}, \mathbf{q}, \mathbf{q}} . \end{aligned} \quad (3.8)$$

The operators acting on the states being decoupled contract to form a number operator $n_{\mathbf{q}, \sigma}$ which projects the sum over $\mathbf{q}$ into the states that are initially occupied (particle sector, PS).

The second such contribution is obtained by flipping the sequence of scattering processes:

$$\begin{aligned} T_{P3} G_W T_{P1}^\dagger &= \sum_{\sigma} \sum_{\mathbf{k}_1, \mathbf{k}_2, \mathbf{k}_3, \mathbf{k}_4} \sum_{\mathbf{q}} W_{\mathbf{k}_1, \mathbf{q}, \mathbf{q}, \mathbf{q}} c_{\mathbf{k}_1, \sigma}^\dagger c_{\mathbf{q}, \sigma} n_{\mathbf{q}, \sigma} G_W W_{\mathbf{q}, \mathbf{k}_2, \mathbf{k}_3, \mathbf{k}_4} c_{\mathbf{q}, \sigma}^\dagger c_{\mathbf{k}_2, \sigma} c_{\mathbf{k}_3, \sigma}^\dagger c_{\mathbf{k}_4, \sigma} \\ &= \sum_{\sigma} \sum_{\mathbf{k}_1, \mathbf{k}_2, \mathbf{k}_3, \mathbf{k}_4} c_{\mathbf{k}_1, \sigma}^\dagger c_{\mathbf{k}_2, \sigma} c_{\mathbf{k}_3, \sigma}^\dagger c_{\mathbf{k}_4, \sigma} \sum_{\mathbf{q} \in \text{HS}} W_{\mathbf{q}, \mathbf{k}_2, \mathbf{k}_3, \mathbf{k}_4} G_W W_{\mathbf{k}_1, \mathbf{q}, \mathbf{q}, \mathbf{q}} . \end{aligned} \quad (3.9)$$

By virtue of eq. 3.6, the product of couplings $W_{\mathbf{q}, \mathbf{k}_2, \mathbf{k}_3, \mathbf{k}_4} G_W W_{\mathbf{k}_1, \mathbf{q}, \mathbf{q}, \mathbf{q}}$ is the same irrespective of whether $\mathbf{q}$ belongs to the particle or hole sector. The two contributions therefore cancel each other. Moreover, the remaining contributions $T_{P3}^\dagger G_W T_{P1}$ and $T_{P1} G_W T_{P2}^\dagger$ are effectively hermitian conjugates of the two contributions considered above, and therefore also cancel each other.

Scattering arising from spin-flip processes

We now come to the processes that involve spin-flips. Considering T_{F1} and T_{F4} first, we get

$$\begin{aligned} T_{F1}^\dagger G_W T_{F4} &= \sum_{\sigma} \sum_{\mathbf{k}_1, \mathbf{k}_2, \mathbf{k}_3, \mathbf{k}_4} \sum_{\mathbf{q}} W_{\mathbf{q}, \mathbf{k}_2, \mathbf{k}_3, \mathbf{k}_4} c_{\mathbf{q}, \sigma}^\dagger c_{\mathbf{k}_2, \sigma} c_{\mathbf{k}_3, \bar{\sigma}}^\dagger c_{\mathbf{k}_4, \bar{\sigma}} G_W W_{\mathbf{k}_1, \mathbf{q}, \mathbf{q}, \mathbf{q}} c_{\mathbf{k}_1, \sigma}^\dagger c_{\mathbf{q}, \sigma} n_{\mathbf{q}, \bar{\sigma}} \\ &= - \sum_{1,2,3,4} \sum_{\sigma} c_{\mathbf{k}_1, \sigma}^\dagger c_{\mathbf{k}_2, \sigma} c_{\mathbf{k}_3, \bar{\sigma}}^\dagger c_{\mathbf{k}_4, \bar{\sigma}} \sum_{\mathbf{q} \in \text{PS}} W_{\mathbf{q}, \mathbf{k}_2, \mathbf{k}_3, \mathbf{k}_4} G_W W_{\mathbf{k}_1, \mathbf{q}, \mathbf{q}, \mathbf{q}} , \\ T_{F4} G_W T_{F1}^\dagger &= \sum_{\sigma} \sum_{\mathbf{k}_1, \mathbf{k}_2, \mathbf{k}_3, \mathbf{k}_4} \sum_{\mathbf{q}} W_{\mathbf{k}_1, \mathbf{q}, \mathbf{q}, \mathbf{q}} c_{\mathbf{k}_1, \sigma}^\dagger c_{\mathbf{q}, \sigma} n_{\mathbf{q}, \bar{\sigma}} G_W W_{\mathbf{q}, \mathbf{k}_2, \mathbf{k}_3, \mathbf{k}_4} c_{\mathbf{q}, \sigma}^\dagger c_{\mathbf{k}_2, \sigma} c_{\mathbf{k}_3, \bar{\sigma}}^\dagger c_{\mathbf{k}_4, \bar{\sigma}} \\ &= \sum_{1,2,3,4} \sum_{\sigma} c_{\mathbf{k}_1, \sigma}^\dagger c_{\mathbf{k}_2, \sigma} c_{\mathbf{k}_3, \bar{\sigma}}^\dagger c_{\mathbf{k}_4, \bar{\sigma}} \sum_{\mathbf{q} \in \text{HS}} W_{\mathbf{q}, \mathbf{k}_2, \mathbf{k}_3, \mathbf{k}_4} G_W W_{\mathbf{k}_1, \mathbf{q}, \mathbf{q}, \mathbf{q}} . \end{aligned} \quad (3.10)$$

By the same arguments as in the previous subsection, these terms cancel each other out. Their hermitian conjugate contributions $T_{F1}G_W T_{F4}^\dagger$ and $T_{F4}^\dagger G_W T_{F1}$ also cancel out. The other two terms are T_{F2} and T_{F3} , and their contributions also cancel out for the same reason:

$$\begin{aligned}
 T_{F2}G_W T_{F2} &= \sum_{\sigma} \sum_{\mathbf{k}_1, \mathbf{k}_2, \mathbf{k}_3, \mathbf{k}_4} \sum_{\mathbf{q}} W_{\mathbf{q}, \bar{\mathbf{q}}, \mathbf{k}_3, \mathbf{k}_4} c_{\mathbf{q}, \sigma}^\dagger c_{\bar{\mathbf{q}}, \sigma} c_{\mathbf{k}_3, \bar{\sigma}}^\dagger c_{\mathbf{k}_4, \bar{\sigma}} G_W W_{\bar{\mathbf{q}}, \mathbf{q}, \mathbf{k}_1, \mathbf{k}_2} c_{\bar{\mathbf{q}}, \sigma}^\dagger c_{\mathbf{q}, \sigma} c_{\mathbf{k}_1, \bar{\sigma}}^\dagger c_{\mathbf{k}_2, \bar{\sigma}} \\
 &= \sum_{1,2,3,4} \sum_{\sigma} c_{\mathbf{k}_1, \sigma}^\dagger c_{\mathbf{k}_2, \sigma} c_{\mathbf{k}_3, \bar{\sigma}}^\dagger c_{\mathbf{k}_4, \bar{\sigma}} \sum_{\mathbf{q} \in \text{PS}} W_{\mathbf{q}, \bar{\mathbf{q}}, \mathbf{k}_3, \mathbf{k}_4} G_W W_{\bar{\mathbf{q}}, \mathbf{q}, \mathbf{k}_1, \mathbf{k}_2} , \\
 T_{F3}G_W T_{F3} &= \sum_{\sigma} \sum_{\mathbf{k}_1, \mathbf{k}_2, \mathbf{k}_3, \mathbf{k}_4} \sum_{\mathbf{q}} W_{\mathbf{q}, \mathbf{k}_2, \mathbf{k}_3, \bar{\mathbf{q}}} c_{\mathbf{q}, \sigma}^\dagger c_{\mathbf{k}_2, \sigma} c_{\mathbf{k}_3, \bar{\sigma}}^\dagger c_{\bar{\mathbf{q}}, \bar{\sigma}} G_W W_{\bar{\mathbf{q}}, \mathbf{k}_4, \mathbf{k}_1, \mathbf{q}} c_{\bar{\mathbf{q}}, \bar{\sigma}}^\dagger c_{\mathbf{k}_4, \bar{\sigma}} c_{\mathbf{k}_1, \sigma}^\dagger c_{\mathbf{q}, \sigma} \\
 &= - \sum_{1,2,3,4} \sum_{\sigma} c_{\mathbf{k}_1, \sigma}^\dagger c_{\mathbf{k}_2, \sigma} c_{\mathbf{k}_3, \bar{\sigma}}^\dagger c_{\mathbf{k}_4, \bar{\sigma}} \sum_{\mathbf{q} \in \text{PS}} W_{\mathbf{q}, \mathbf{k}_2, \mathbf{k}_3, \bar{\mathbf{q}}} G_W W_{\bar{\mathbf{q}}, \mathbf{k}_4, \mathbf{k}_1, \mathbf{q}} ,
 \end{aligned} \tag{3.11}$$

Scattering involving both spin-flip and spin-preserving processes

These processes involve the combination of terms like T_{P1} with T_{F4} , and T_{P2} with T_{F1} . These again cancel each other out for the same reasons as outline above.

Net renormalisation for the bath correlation term

Since all the contributions cancel out in pairs, the bath correlation term W is *marginal*.

1.3.3 Renormalisation of the Kondo scattering term J

We focus on the renormalisation of the spin-flip part of the Kondo interaction. For these processes, the intermediate many-body state always involves the impurity spin being anti-correlated with the conduction electron spin, such that the propagator for that state is $G_J = 1/(\omega - |\varepsilon_j|/2 + J_{\mathbf{q}}/4 + W_{\mathbf{q}}/2)$.

Impurity-mediated spin-flip scattering purely through Kondo-like processes

The following processes arising from the Kondo term renormalise the spin-flip interaction:

$$\begin{aligned}
 T_{KT1}^\dagger G_J (T_{KZ1} + T_{KZ1}^\dagger) &= \frac{1}{4} \sum_{\mathbf{k}_1, \mathbf{k}_1, \mathbf{q}} J_{\mathbf{q}, \mathbf{k}_2} S_d^+ \left[-c_{\mathbf{q}\downarrow}^\dagger c_{\mathbf{k}_2\uparrow} G_J c_{\mathbf{k}_1\downarrow}^\dagger c_{\mathbf{q}\downarrow} + c_{\mathbf{k}_2\downarrow}^\dagger c_{\mathbf{q}\uparrow} G_J c_{\mathbf{q}\uparrow}^\dagger c_{\mathbf{k}_1\uparrow} \right] J_{\mathbf{k}_1, \mathbf{q}} S_d^z \\
 &= -\frac{1}{8} \sum_{\mathbf{k}_1, \mathbf{k}_1, \mathbf{q}} J_{\mathbf{q}, \mathbf{k}_2} S_d^+ \left[c_{\mathbf{k}_1\downarrow}^\dagger c_{\mathbf{k}_2\uparrow} G_J n_{\mathbf{q}\downarrow} + c_{\mathbf{k}_2\downarrow}^\dagger c_{\mathbf{k}_1\uparrow} (1 - n_{\mathbf{q}\uparrow}) G_J \right] J_{\mathbf{k}_1, \mathbf{q}} \\
 &= -\frac{1}{8} \sum_{\mathbf{k}_1, \mathbf{k}_1} S_d^+ c_{\mathbf{k}_1\downarrow}^\dagger c_{\mathbf{k}_2\uparrow} \sum_{\mathbf{q} \in \text{PS}} [J_{\mathbf{q}, \mathbf{k}_2} J_{\mathbf{k}_1, \mathbf{q}} + J_{\bar{\mathbf{q}}, \mathbf{k}_1} J_{\mathbf{k}_2, \bar{\mathbf{q}}}] G_J .
 \end{aligned} \tag{3.12}$$

In getting the final expression, we used the sigma matrix relation $S_d^+ S_d^z = -\frac{1}{2} S_d^+$, and absorbed the projector $1 - n_{\mathbf{q}\uparrow}$ into the sum over the particle sector by replacing q with its particle-hole transformed

counterpart $\bar{q}$. An identical contribution is obtained by switching the sequence of processes:

$$\begin{aligned} (T_{KZ1} + T_{KZ1}^\dagger) G_J T_{KT1}^\dagger &= \frac{1}{4} \sum_{\mathbf{k}_1, \mathbf{k}_1, \mathbf{q}} J_{\mathbf{k}_1, \mathbf{q}} S_d^z \left[-c_{\mathbf{k}_1 \downarrow}^\dagger c_{\mathbf{q} \downarrow} G_J c_{\mathbf{q} \downarrow}^\dagger c_{\mathbf{k}_2 \uparrow} + c_{\mathbf{q} \uparrow}^\dagger c_{\mathbf{k}_1 \uparrow} G_J c_{\mathbf{k}_2 \downarrow}^\dagger c_{\mathbf{q} \uparrow} \right] J_{\mathbf{q}, \mathbf{k}_2} S_d^+ \\ &= -\frac{1}{8} \sum_{\mathbf{k}_1, \mathbf{k}_1} S_d^+ c_{\mathbf{k}_1 \downarrow}^\dagger c_{\mathbf{k}_2 \uparrow} \sum_{\mathbf{q} \in \text{PS}} [J_{\bar{\mathbf{q}}, \mathbf{k}_2} J_{\mathbf{k}_1, \bar{\mathbf{q}}} + J_{\mathbf{q}, \mathbf{k}_1} J_{\mathbf{k}_2, \mathbf{q}}] G_J. \end{aligned} \quad (3.13)$$

Scattering processes involving interplay between the Kondo interaction and conduction bath interaction

Looking at $T_{KT1}^\dagger$ first, we have

$$T_{KT1}^\dagger G_J (T_{F4} + T_{F4}^\dagger) = \frac{1}{2} \sum_{\mathbf{k}_1, \mathbf{k}_2, \mathbf{q}} J_{\mathbf{k}_2, \mathbf{q}} S_d^+ \left(c_{\mathbf{q} \downarrow}^\dagger c_{\mathbf{k}_2 \uparrow} G_J W_{\mathbf{q}, \mathbf{q}, \mathbf{k}_1, \mathbf{q}} n_{\mathbf{q} \uparrow} c_{\mathbf{k}_1 \downarrow}^\dagger c_{\mathbf{q} \downarrow} + c_{\mathbf{k}_2 \downarrow}^\dagger c_{\mathbf{q} \uparrow} G_J W_{\bar{\mathbf{q}}, \bar{\mathbf{q}}, \mathbf{q}, \mathbf{k}_1} n_{\bar{\mathbf{q}} \downarrow} c_{\mathbf{q} \uparrow}^\dagger c_{\mathbf{k}_1 \uparrow} \right). \quad (3.14)$$

For either of the two choices of the functional form of W , it is easy to show that $W_{\mathbf{q}, \mathbf{q}, \mathbf{k}_1, \mathbf{q}} = W_{\bar{\mathbf{q}}, \bar{\mathbf{q}}, \mathbf{q}, \mathbf{k}_1}$.

$$T_{KT1}^\dagger G_J (T_{F4} + T_{F4}^\dagger) = \frac{1}{2} \sum_{\mathbf{k}_1, \mathbf{k}_2, \mathbf{q}} J_{\mathbf{k}_2, \mathbf{q}} W_{\mathbf{q}, \mathbf{q}, \mathbf{k}_1, \mathbf{q}} G_J S_d^+ \left[-c_{\mathbf{k}_1 \downarrow}^\dagger c_{\mathbf{k}_2 \uparrow} n_{\mathbf{q} \downarrow} n_{\mathbf{q} \uparrow} + c_{\mathbf{k}_2 \downarrow}^\dagger c_{\mathbf{k}_1 \uparrow} (1 - n_{\mathbf{q} \uparrow}) n_{\bar{\mathbf{q}} \downarrow} \right]. \quad (3.15)$$

Another contribution is obtained by switching the sequence of the scattering processes:

$$\begin{aligned} (T_{F4} + T_{F4}^\dagger) G_J T_{KT1}^\dagger &= \frac{1}{2} \sum_{\mathbf{k}_1, \mathbf{k}_2, \mathbf{q}} \left(W_{\mathbf{q}, \mathbf{q}, \mathbf{k}_1, \mathbf{q}} n_{\bar{\mathbf{q}} \uparrow} c_{\mathbf{k}_1 \downarrow}^\dagger c_{\mathbf{q} \downarrow} G_J c_{\mathbf{q} \downarrow}^\dagger c_{\mathbf{k}_2 \uparrow} + W_{\bar{\mathbf{q}}, \bar{\mathbf{q}}, \mathbf{q}, \mathbf{k}_1} n_{\mathbf{q} \downarrow} c_{\mathbf{q} \uparrow}^\dagger c_{\mathbf{k}_1 \uparrow} G_J c_{\mathbf{k}_2 \downarrow}^\dagger c_{\mathbf{q} \uparrow} \right) J_{\mathbf{k}_2, \mathbf{q}} S_d^+ \\ &= \frac{1}{2} \sum_{\mathbf{k}_1, \mathbf{k}_2, \mathbf{q}} \left(c_{\mathbf{k}_1 \downarrow}^\dagger c_{\mathbf{k}_2 \uparrow} n_{\bar{\mathbf{q}} \uparrow} (1 - n_{\mathbf{q} \downarrow}) - c_{\mathbf{k}_2 \downarrow}^\dagger c_{\mathbf{k}_1 \uparrow} n_{\mathbf{q} \downarrow} n_{\mathbf{q} \uparrow} \right) W_{\mathbf{q}, \mathbf{q}, \mathbf{k}_1, \mathbf{q}} G_J J_{\mathbf{k}_2, \mathbf{q}} S_d^+ \end{aligned} \quad (3.16)$$

The two contributions (eqs. 3.15 and 3.16) arising from T_{KT1} cancel each other.

We now consider the other spin-exchange process $T_{KT2}^\dagger$. One such contribution is

$$\begin{aligned} T_{KT2}^\dagger G_J T_{F3} &= \frac{1}{2} \sum_{\mathbf{k}_1, \mathbf{k}_2, \mathbf{q}} J_{\mathbf{q}, \bar{\mathbf{q}}} S_d^+ \left(c_{\mathbf{q} \downarrow}^\dagger c_{\bar{\mathbf{q}} \uparrow} G_J c_{\bar{\mathbf{q}} \uparrow}^\dagger c_{\mathbf{k}_2 \uparrow} c_{\mathbf{k}_1 \downarrow}^\dagger c_{\mathbf{q} \downarrow} + c_{\bar{\mathbf{q}} \downarrow}^\dagger c_{\mathbf{q} \uparrow} G_J c_{\mathbf{q} \uparrow}^\dagger c_{\mathbf{k}_2 \uparrow} c_{\mathbf{k}_1 \downarrow}^\dagger c_{\bar{\mathbf{q}} \downarrow} \right) W_{\bar{\mathbf{q}}, \mathbf{k}_2, \mathbf{k}_1, \mathbf{q}} \\ &= -\frac{1}{2} \sum_{\mathbf{k}_1, \mathbf{k}_2, \mathbf{q}} S_d^+ c_{\mathbf{k}_1 \downarrow}^\dagger c_{\mathbf{k}_2 \uparrow} \left[n_{\mathbf{q} \downarrow} (1 - n_{\bar{\mathbf{q}} \uparrow}) + n_{\bar{\mathbf{q}} \downarrow} (1 - n_{\mathbf{q} \uparrow}) \right] J_{\mathbf{q}, \bar{\mathbf{q}}} G_J W_{\bar{\mathbf{q}}, \mathbf{k}_2, \mathbf{k}_1, \mathbf{q}} \\ &= -\frac{1}{2} \sum_{\mathbf{k}_1, \mathbf{k}_2} S_d^+ c_{\mathbf{k}_1 \downarrow}^\dagger c_{\mathbf{k}_2 \uparrow} \sum_{\mathbf{q} \in \text{PS}} (J_{\mathbf{q}, \bar{\mathbf{q}}} W_{\bar{\mathbf{q}}, \mathbf{k}_2, \mathbf{k}_1, \mathbf{q}} + J_{\bar{\mathbf{q}}, \mathbf{q}} W_{\mathbf{q}, \mathbf{k}_2, \mathbf{k}_1, \bar{\mathbf{q}}}) G_J. \end{aligned} \quad (3.17)$$

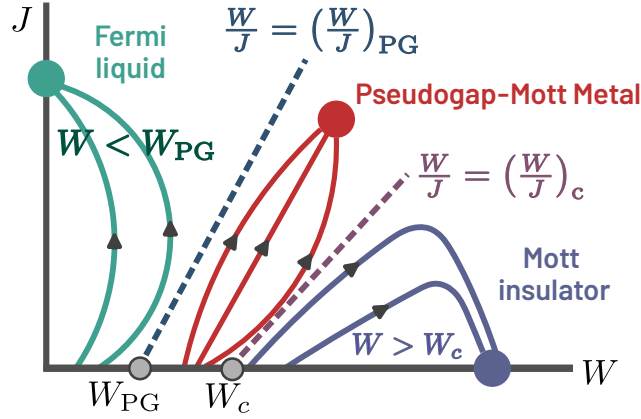


Figure 1.4: Renormalisation group (RG) flow diagram of the lattice model as obtained from a unitary RG analysis of the lattice-embedded impurity model. For $|W/J| < |W/J|_{PG}$, the flows lead to a Fermi liquid phase, while for $|W/J| < |W/J|_c$, they lead to a Mott insulator. For values in between, the system flows to a Mott metal phase that has a pseudogap in the density of states.

An identical contribution is obtained from the reversed term:

$$\begin{aligned}
 T_{F3} G_J T_{KT2}^\dagger &= \frac{1}{2} \sum_{\mathbf{k}_1, \mathbf{k}_2, \mathbf{q}} W_{\bar{\mathbf{q}}, \mathbf{k}_2, \mathbf{k}_1, \mathbf{q}} \left(c_{\bar{\mathbf{q}}\uparrow}^\dagger c_{\mathbf{k}_2\uparrow} c_{\mathbf{k}_1\downarrow}^\dagger c_{\mathbf{q}\downarrow} G_J c_{\mathbf{q}\downarrow}^\dagger c_{\bar{\mathbf{q}}\uparrow} + c_{\mathbf{q}\uparrow}^\dagger c_{\mathbf{k}_2\uparrow} c_{\mathbf{k}_1\downarrow}^\dagger c_{\bar{\mathbf{q}}\downarrow} G_J c_{\bar{\mathbf{q}}\downarrow}^\dagger c_{\mathbf{q}\uparrow} \right) J_{\mathbf{q}, \bar{\mathbf{q}}} S_d^+ \\
 &= -\frac{1}{2} \sum_{\mathbf{k}_1, \mathbf{k}_2} S_d^+ c_{\mathbf{k}_1\downarrow}^\dagger c_{\mathbf{k}_2\uparrow} \sum_{\mathbf{q} \in \text{PS}} (J_{\mathbf{q}, \bar{\mathbf{q}}} W_{\bar{\mathbf{q}}, \mathbf{k}_2, \mathbf{k}_1, \mathbf{q}} + J_{\bar{\mathbf{q}}, \mathbf{q}} W_{\mathbf{q}, \mathbf{k}_2, \mathbf{k}_1, \bar{\mathbf{q}}}) G_J.
 \end{aligned} \tag{3.18}$$

Net renormalisation to the Kondo interaction

Combining the results from eqs. 3.12, 3.13, 3.17 and 3.18, as well as using the properties $J_{\bar{\mathbf{q}}, \mathbf{k}_1} J_{\mathbf{k}_2, \bar{\mathbf{q}}} = J_{\mathbf{q}, \mathbf{k}_2} J_{\mathbf{k}_1, \mathbf{q}} = J_{\mathbf{k}_2, \mathbf{q}} J_{\mathbf{q}, \mathbf{k}_1}$ and $J_{\mathbf{q}, \bar{\mathbf{q}}} W_{\bar{\mathbf{q}}, \mathbf{k}_2, \mathbf{k}_1, \mathbf{q}} = J_{\bar{\mathbf{q}}, \mathbf{q}} W_{\mathbf{q}, \mathbf{k}_2, \mathbf{k}_1, \bar{\mathbf{q}}}$, the total renormalisation in the momentum-resolved Kondo coupling $J^{(j)}$ at the j^{th} step amounts to

$$\Delta J_{\mathbf{k}_1, \mathbf{k}_2}^{(j)} = - \sum_{\mathbf{q} \in \text{PS}} \frac{J_{\mathbf{k}_2, \mathbf{q}}^{(j)} J_{\mathbf{q}, \mathbf{k}_1}^{(j)} + 4 J_{\mathbf{q}, \bar{\mathbf{q}}}^{(j)} W_{\bar{\mathbf{q}}, \mathbf{k}_2, \mathbf{k}_1, \mathbf{q}}}{\omega - \frac{1}{2} |\varepsilon_j| + J_{\mathbf{q}}^{(j)} / 4 + W_{\mathbf{q}} / 2} \tag{3.19}$$

A representation of the RG phase diagram in the $-W/t$ vs. J/t plane of couplings has been provided in Fig. 1.5(a) from detailed numerical computations of eq.(??). We also provide a schematic representation of RG flows to various fixed point theories in the J vs. W plane of couplings in Fig. 1.4. This schematic figure shows the existence of three distinct phases (Fermi liquid metal, Mott Insulator, and Pseudogap-Mott metal) described by distinct fixed point theories of the RG flows. The Fermi liquid metal and Mott Insulator phases are distinguished from the Pseudogap-Mott metal by separatrices of the RG flows.

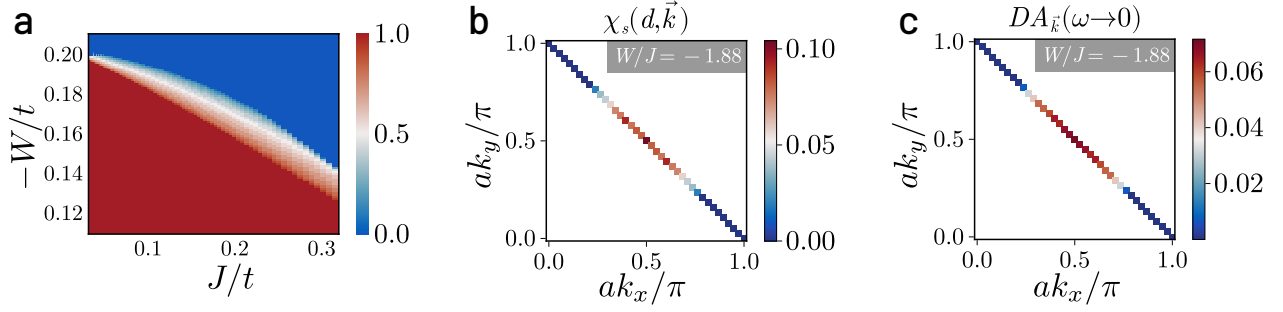


Figure 1.5: (a) Phase diagram of impurity model at strong coupling in U in terms of competing dimensionless Kondo (J/t) and bath correlation (W/t) couplings. Colourbar represents the fraction of states on the Fermi surface replaced by zeros of the Greens function (Luttinger surface). A pseudogap phase (shaded region) is observed between the local Fermi liquid (red region) and local moment (blue region) phases (b) k -space-resolved spin-spin correlation $\chi_s(d, \mathbf{k}) = \langle \mathbf{S}_d \cdot \mathbf{S}_{\mathbf{k}} \rangle$ in the pseudogap phase of the impurity model. Antinodal regions are observed to decouple from Kondo screening of the impurity. (c) Upon tiling, this leads to a k -space-resolved antinodal gap in the electronic density of states of the lattice model, corresponding to Luttinger surfaces of zeros.

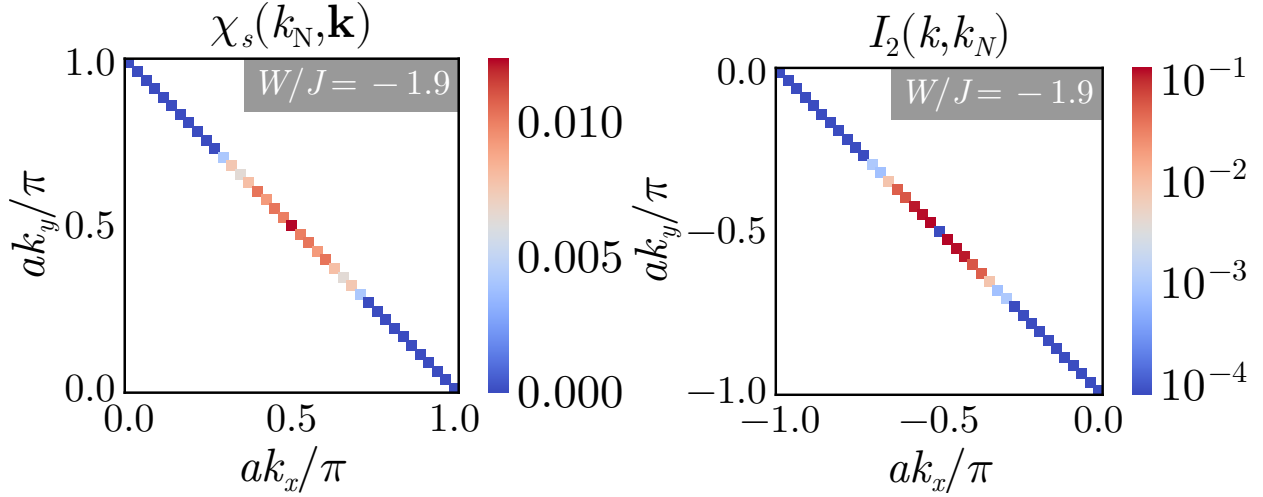


Figure 1.6: Left: Spin correlations χ_s for the tiled model, between the nodal point $k_N = (-\pi/2, -\pi/2)$ and an arbitrary k -point on the Fermi surface. In the PG, the spin-correlations near the antinode vanish, indicating that they have been removed from the metallic excitations, while the correlations near the nodal point become enhanced because of the increasingly correlated nature of the metal. Right: Mutual information I_2 between an arbitrary k -state and the nodal point. The nodal points remain strongly entangled with each other all the way through the PG.

1.4 Reconstructing of lattice model from auxiliary model

To reconstruct a translationally invariant lattice Hamiltonian from the impurity model (with Hamiltonian $\mathcal{H}_{\text{aux}}$), we define a many-body tiling prescription that embeds the auxiliary impurity system uniformly across the two-dimensional square lattice. Thus, the tiling process systematically derives

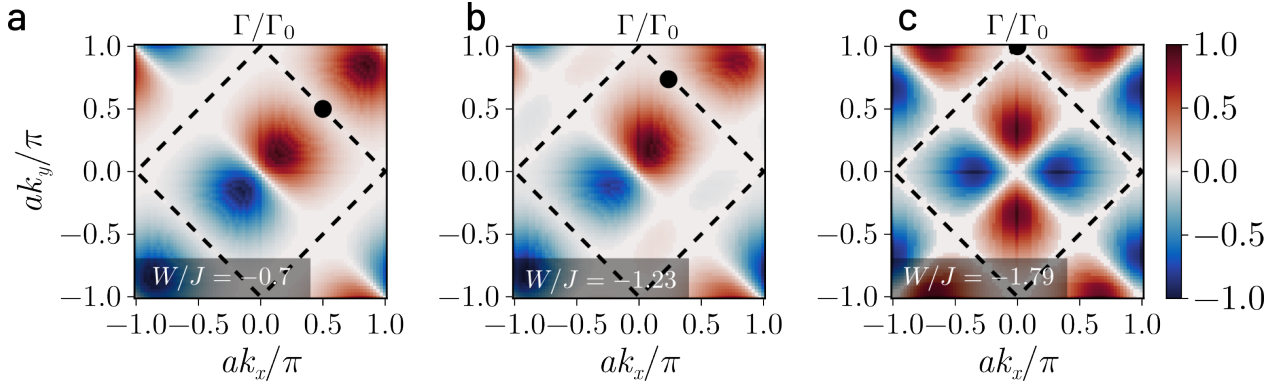


Figure 1.7: Initiation of the decoupling of $J_{\mathbf{k},\mathbf{k}'}$ (positive, negative and zeros shown in red, blue and white respectively), with $\mathbf{k}$ (black circle) at low-energies for $\mathbf{k}$ corresponding to the node (a), antinode (b) and a point mid-way between them (c) on the top right arm of the FS respectively with tuning W/J . As dictated by the symmetry of $J_{\mathbf{k},\mathbf{k}'}$, the decoupling for a given $\mathbf{k}$ proceeds via the appearance of zeros (white patches) of $J_{\mathbf{k},\mathbf{k}'}$ for $\mathbf{k}'$ initially on the nodal regions of adjacent arms, and progresses gradually towards the antinodes. The decoupling ends with the onset of the pseudogap.

the full lattice dynamics from the renormalized fixed-point structure of the impurity. As described in Section 1.2, the auxiliary model consists of a single impurity embedded in a correlated bath at the location $\mathbf{r}_d$:

$$H(\mathbf{r}_d) = H_{\text{imp}} + H_{\text{cbath}} + H_{\text{imp-cbath}}. \quad (4.1)$$

The form of various terms in eq.(4.1) have been defined in eqs.(2.2), (2.3) and (2.4) in Section 1.2. The tiling procedure is shown in detail in Section ??.

A non-zero chemical potential in the bath and/or the impurity can be included to study the effects of doping. Crucially, the eigenstates of $\mathcal{H}_{\text{tilde}}$ obey a many-body generalization of Bloch's theorem [57], which allows for the exact computation of momentum-resolved observables (presented here for a 77×77 $\mathbf{k}$ -space Brillouin zone grid). We shall demonstrate below that the lattice embedding and tiling procedures provide a controlled route to capture low-energy NFL behavior and Mott criticality directly from a quantum impurity model.

1.5 Low-energy theory for the lattice-embedded impurity model

1.5.1 RG Phase Diagram

A detailed picture of the PG in the impurity model is obtained from a momentum-resolved breakdown of Kondo screening in the strong-coupling regime $U \gg t$ phase of H_{aux} . A unitary renormalisation group (URG) scaling analysis [44] obtains a flow equation of the Kondo coupling $J_{\mathbf{k}_1,\mathbf{k}_2}^{(j)}$ (see

Appendix A for details)

$$\Delta J_{\mathbf{k}_1, \mathbf{k}_2}^{(j)} = - \sum_{\mathbf{q} \in \text{PS}} \frac{J_{\mathbf{k}_2, \mathbf{q}}^{(j)} J_{\mathbf{q}, \mathbf{k}_1}^{(j)} + 4J_{\mathbf{q}, \bar{\mathbf{q}}}^{(j)} W_{\bar{\mathbf{q}}, \mathbf{k}_2, \mathbf{k}_1, \mathbf{q}}}{\omega - \frac{1}{2}|\varepsilon_j| + J_{\mathbf{q}}^{(j)}/4 + W_{\mathbf{q}}/2}, \quad (5.1)$$

where ε_j is the energy of the shell being decoupled at the j^{th} step, the sum is over all occupied momentum states $\mathbf{q}$ of the energy shell ε_j , and $\bar{\mathbf{q}} = \mathbf{q} + \boldsymbol{\pi}$ is the particle-hole transformed state associated with $\mathbf{q}$. The bath interaction coupling $W_{\bar{\mathbf{q}}, \mathbf{k}_2, \mathbf{k}_1, \mathbf{q}}$ is found to be marginal under these transformations. While we present a detailed numerical evaluation of the RG equation for $J_{\mathbf{k}_1, \mathbf{k}_2}$ (eq.(5.1)) below, it is clear that the frustration of Kondo screening due to charge fluctuations (for attractive bath interactions $W < 0$) leads to the Mott transition [58].

A detailed numerical computation of the URG equation in eq. (5.1) leads to the phase diagram shown in Fig. 1.5(a). Upon tuning the ratio of the bath and Kondo interactions (W/J) from zero to negative values, the following phases emerge in the impurity model from the competition between J and W in eq. (5.1): (i) for $W/J < (W/J)_{\text{PG}}$, an LFL phase (red region), where the entire FS participates in Kondo screening, (ii) for $\frac{W}{J} \in [(\frac{W}{J})_{\text{PG}}, (\frac{W}{J})_c]$ (shaded region), a local PG phase where disconnected parts of the FS around the node participate in Kondo screening, and (iii) a local moment phase for $\frac{W}{J} > (\frac{W}{J})_c$ (blue region), where the impurity remains unscreened at low-energies. These can be visualised from spin correlations, $\chi_s(d, \mathbf{k}) = \langle \mathbf{S}_d \cdot \mathbf{S}_{\mathbf{k}} \rangle$, as shown in Fig. 1.5(b) for the PG. The values $(W/J)_{\text{PG}}$ and $(W/J)_c$ are therefore the entry into and exit from the PG phase.

Mapping onto the lattice model via tiling, we observe that RG-induced nodal–antinodal dichotomy in $J_{\mathbf{k}, \mathbf{k}'}$ in the impurity model is the microscopic origin of the PG in the lattice model: the extinction of Kondo coherence translates into spectral zeros in the lattice Green’s function (i.e., Luttinger surfaces) at antinodal momenta (Fig. 1.5(c)). This establishes that the $T = 0$ Mott transition of the 2D *extended*-Hubbard model proceeds from FL to MI through an intervening PG phase. In Fig. 1.6, similar insights obtain from the spin-correlation $\chi_s(\mathbf{k}, k_N) = \langle \mathbf{S}_{\mathbf{k}} \cdot \mathbf{S}_{k_N} \rangle$ in k -space (where k_N is the nodal momentum) and mutual information $I_2(\mathbf{k} : k_N)$ between two momentum states: the PG displays Fermi arcs in the nodal regions along with vanishing correlation and entanglement in the antinodal regions.

1.5.2 Unravelling of Kondo Screening

The $\mathbf{k}$ -space anisotropy of Kondo breakdown can be visualized in terms of zeros of $J_{\mathbf{k}_N, \mathbf{k}}$, involving spin-flip scattering between the node $\mathbf{k}_N = (\pi/2, \pi/2)$ and a general wavevector $\mathbf{k}$. For any W/J , the C_4 lattice symmetry dictates that $J_{\mathbf{k}_N, \mathbf{k}}$ vanishes if $\mathbf{k}$ belongs to any of the antinodes or adjacent nodes. Tuning W/J towards $(W/J)_{\text{PG}}$ leads to an unravelling of the Kondo screening: $J_{\mathbf{k}_N, \mathbf{k}}$ for $\mathbf{k}$ close to the adjacent nodes turns RG-irrelevant first, and a patch of zeros subsequently appears in $J_{\mathbf{k}_N, \mathbf{k}}$ around this point (Fig. 1.7(a)). Tuning W/J further extends the patch of zeros towards the antinodes (Fig. 1.7(b) and 1.7(c)). Kondo screening thus unravels by a systematic decoupling of all $J_{\mathbf{k}_1, \mathbf{k}_2}$ that connect adjacent quadrants of the Brillouin zone.

Precisely at $W/J = (W/J)_{\text{PG}}$, the antinode joins this connected region of zeros in $J_{\mathbf{k}_1, \mathbf{k}_2}$, marking the decoupling of the antinodes from all other points in the neighbourhood of the FS. This is an interaction-driven Lifshitz transition of the FS, and marks the entry into a PG phase possessing

Fermi arcs [59]. Importantly, it coincides with an emergent two-channel Kondo (2CK) impurity model, where each channel corresponds to a pair of Fermi arcs on opposite faces of the conduction bath FS (see Subection 1.5.3). The 2CK nature of the PG is guaranteed by the symmetry of $J_{\mathbf{k},\mathbf{k}'}$:

$$J_{\mathbf{k},\mathbf{k}'} = -J_{\mathbf{k}+\mathbf{Q},\mathbf{k}'} = -J_{\mathbf{k},\mathbf{k}'+\mathbf{Q}}, \quad \mathbf{Q} = (\pi, \pi). \quad (5.2)$$

The PG expands by shrinking these disconnected Fermi arcs towards the respective nodes, leading to nodal metals whose disappearance heralds the Mott transition.

1.5.3 Emergence of a two-channel Kondo Model

As the bath interaction W is tuned through the L-PG phase, the four nodal points in the Brillouin zone are the last to decouple from the impurity. This allows us to write down a simpler Kondo model near the transition, focusing on scattering processes involving the nodal region. Specifically, for the Kondo term $J_{\mathbf{k}_1,\mathbf{k}_2}$, we find that only the following classes of scattering processes survive close to the critical point:

- Both momenta in a neighbourhood around $\mathbf{k}_N$, where $\mathbf{k}_N$ can be any of the four nodal points,
- One momentum from the neighbourhood around $\mathbf{k}_N$, while the other from the neighbourhood around $\mathbf{k}_N + (\pi, \pi)$.

The second class is tied to the first through a symmetry of the Hamiltonian (eq. ??). The crucial feature of these processes is that each node only interacts with the other node directly opposite to it across the origin. That is, $(\pi/2, \pi/2)$ and $(-\pi/2, -\pi/2)$ only interact between themselves, while $(\pi/2, -\pi/2)$ and $(-\pi/2, \pi/2)$ form their own pair of connected regions.

Let $S^\pm$ be the set of momentum states in a small window around nodes along $k_x = \pm k_y$. As discussed immediately above, the Kondo spin-exchange term close to the transition does not lead to scattering of any k -state from S^+ to S^- . This is verified by calculating the ratio $\max \{J_{k_1^+, k_2^-}\} / \max \{J_{k_1^+, k_2^+}\}$, where the maximum in the numerator is calculated from all the low-energy (fixed point) Kondo scattering processes that connect the two sets $S^\pm$, while the denominator maximum is from within the set S^+ . This ratio vanishes (left panel of Fig. 1.10) within the local pseudogap regime, indicating that no direct Kondo scattering process persists between the two sectors $S^\pm$ at low energies, deep within the pseudogap regime. We call the value of W at which this ratio vanishes W^* .

The decoupling of the k -space regions $S^\pm$ is captured by the following simplified low-energy Hamiltonian:

$$H_{\text{eff}} = H(S^+) + H(S^-); \quad H(S) = \sum_{\mathbf{q} \in S, \sigma} \varepsilon_{\mathbf{q}} c_{\mathbf{q}\sigma}^\dagger c_{\mathbf{q}\sigma} + \sum_{\mathbf{q}_1 \in S, \mathbf{q}_2 \in S} \sum_{\alpha, \beta} J^*(\mathbf{q}_1, \mathbf{q}_2) \mathbf{S}_d \cdot \sigma_{\alpha\beta} c_{\mathbf{q}_1\alpha}^\dagger c_{\mathbf{q}_2\beta}, \quad (5.3)$$

where $H(S^\pm)$ represents the dynamics of momentum states residing in regions $S^\pm$. The only source of information transfer between the two regions is the fact that they interact with the same impurity local moment. Eq. 5.3 tells us that the low-energy dynamics of the embedded eSIAM close to the Kondo-breakdown transition is governed by an emergent **two-channel Kondo model**, with the conduction bath states in the sets $S^\pm$ making up the two channels.

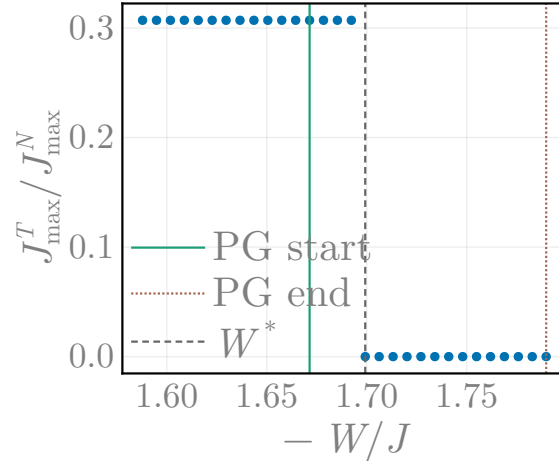


Figure 1.8: Ratio between the maximum values of the fixed point values of the Kondo couplings for scattering processes between and within the regions S^+ and S^- . The vanishing of the fixed point ratio indicates that the pseudogap regime features a singular low-energy effective theory, where the Fermi surface neighbourhood splits into two disconnected regions with no explicit term in the Hamiltonian connecting them directly.

1.5.4 Momentum-resolved Dynamical Spectral Weight Transfer

Passage through the PG phase is accompanied by a highly structured transfer of spectral weight across the FS. Strong charge fluctuations develop between the nodal and antinodal regions of the FS in the PG regime of the impurity model (Fig. 1.9(a)), as captured by the correlator:

$$\chi_c(\mathbf{k}_1, \mathbf{k}_2) = \left\langle c_{\mathbf{k}_1\uparrow}^\dagger c_{\mathbf{k}_1\downarrow}^\dagger c_{\mathbf{k}_2\downarrow} c_{\mathbf{k}_2\uparrow} + \text{h.c.} \right\rangle. \quad (5.4)$$

These fluctuations dynamically redistribute low-energy spectral weight from the antinodes to higher energies, leading to selective gap formation. Accordingly, the Luttinger surfaces of the PG [3] coincides with the appearance of poles of the lattice model self-energy $\Sigma(\mathbf{k}, \omega = 0)$ near the antinodes; these poles approach the nodes on tuning towards the Mott transition (Fig. 1.9(b)). This mirrors the coalescing of finite-frequency poles of the self-energy poles towards zero frequency in the underlying impurity model (Fig. 1.9(c)).

1.6 Non-Fermi liquid excitations within the Pseudogap

1.6.1 Pseudogapped spectral function and singular self-energy

In the PG regime, the nature of gapless Fermi arcs changes dramatically. We have already argued that the low-energy dynamics of these gapless Fermi arcs are governed by an underlying two-channel Kondo (2CK) impurity model [60, 61]. This is consistent with the rapid fall of the impurity quasi-particle residue Z_{imp} (Fig. 1.10(a)) from finite values in the FL phase to vanishingly small values just before the onset of the PG. Inset shows the simultaneous emergence of increasingly uncompensated

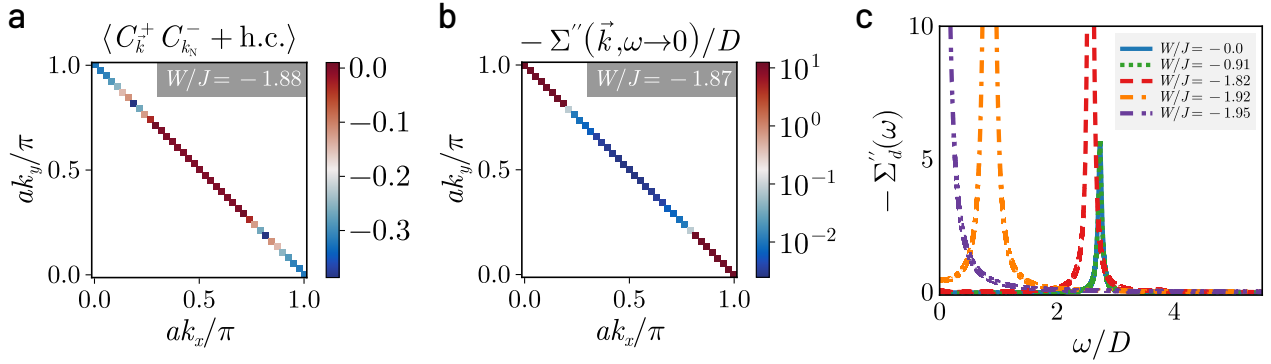


Figure 1.9: (a) Enhanced charge correlations $\chi_c(\mathbf{k}_1, \mathbf{k}_2) = \langle c_{\mathbf{k}_1\uparrow}^\dagger c_{\mathbf{k}_1\downarrow}^\dagger c_{\mathbf{k}_2\downarrow} c_{\mathbf{k}_2\uparrow} + \text{h.c.} \rangle$ between the nodal and antinodal regions, signalling Kondo breakdown in the pseudogap phase of the impurity model. (b) In turn, the breakdown leads to the gapping of the antinodal regions in the lattice model, seen from the appearance of poles in the imaginary part of the self-energy. (c) The imaginary part of the impurity self-energy $\Sigma''(\omega > 0)$ possesses a pole at non-zero ω in the PG that moves towards $\omega = 0$ as the Mott transition is approached.

local magnetic moments upon traversing the PG phase. The accompanying impurity spectral function of the gapless arcs show a pseudogapped behaviour for $\omega \simeq 0$, with a rapid fall in the spectral weight at $\omega = 0$ upon traversing the PG (Fig. 1.10(c)). The collapse of the Kondo resonance into a pseudogapped spectral function is accompanied by the redistribution of spectral weight [3] in the impurity spectral function (Fig. 1.10(b)) from $\omega \sim 0$ to the Hubbard sidebands at finite frequencies $\omega \simeq \pm 3$ (in units of the bandwidth).

Concomitant with this is the emergence of a zero-frequency peak in the imaginary part of the self-energy of the NFL in the PG phase (Fig. 1.11(a)),

$$-1/\Sigma''(\omega) \sim 1/\Sigma''(0) + \omega^\beta. \quad (6.1)$$

Remarkably, we find that the exponent $\beta = 2$ characterises the NFL for the entire PG phase (see Fig. 1.11(b)), including the critical end-point. This is in stark contrast with the $\Sigma''(\omega) \sim \omega^2$ for the FL (Fig. 1.11(b)). A similar analysis reveals the power-law scaling for the low-frequency behaviour of the impurity spectral function in Fig. 1.12. We observe that $A_d(\omega) \sim A_d(0) + \omega^2$ for the pseudogap phase, while $1/A_d(\omega) \sim 1/A_d(0) + \omega^2$ for the Fermi liquid (inset of Fig. 1.12).

1.6.2 Finite-Frequency Scattering Rate and Optical Response Through the Pseudogap

All $\Sigma''(\omega)$ for the NFL are observed to lie above the Mott-Ioffe-Regel (MIR) bound [62, 63] (dashed blue line in Fig. 1.11(a)), while those for the FL lie below. The MIR bound is the maximum expected scattering rate in metals when the mean free path approaches the lattice spacing:

$$-2\Sigma''_{\text{MIR}} = 1/\tau_{\text{MIR}}, \quad \tau_{\text{MIR}} = l_{\text{min}}/v_F, \quad (6.2)$$

where $l_{\min}$ is the minimum mean free path in metals (equal to one lattice spacing) and τ_{MIR} is the associated lifetime of quasiparticles close to the FS (with Fermi velocity v_F). The height of the zero-frequency peak rises by almost 4 orders of magnitude from the start of the PG till its end at the Mott transition point (Fig. 1.11(c)); the dramatic growth of the peak height very near the Mott critical point coincides with the coalescing of the finite-frequency poles of the self-energy into a single pole at zero-frequency, signalling the singular nodal NFL present at the Mott quantum critical point.

To further probe the breakdown of coherent transport in the pseudogap (PG) regime, we examine the product of the impurity spectral function $A(\omega)$ and the scattering lifetime $\tau(\omega)$, which approximates the optical conductivity $\sigma(\omega)$ via a Drude-type relation. As shown in the main panel of Fig. ??, this quantity exhibits a sharp Drude peak at $\omega = 0$ in the Fermi liquid (FL) regime (blue), consistent with long-lived quasiparticles. In stark contrast, the PG regime (green, red, orange) shows a pronounced suppression of low-frequency conductivity and the emergence of a finite-frequency peak centered near $\omega/D \sim 0.3$ – 0.4 (where D is the kinetic energy bandwidth).

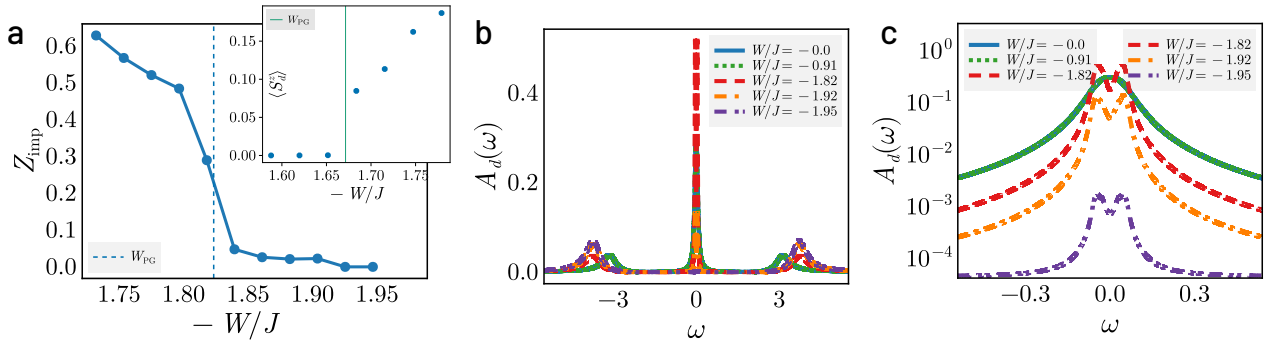


Figure 1.10: (a) Suppression of quasiparticle residue (Z_{imp}) as the impurity model is tuned towards the Mott transition. An initial drastic fall in Z_{imp} is observed for $W/J \lesssim (W/J)_{\text{PG}}$ from 0.3 to around 0.05, signalling the destruction of the FL with unravelling of Kondo screening. A steady decrease in Z_{imp} is observed in passage through the PG, and is vanishingly small close to the Mott transition due to a divergent self-energy. Inset shows the growth of unscreened impurity magnetic moment in the pseudogap phase, signalling the breakdown of Kondo screening. (b) Impurity Spectral function in the FL and PG phases. The evolution of the central peak from Kondo resonance to pseudogap is accompanied by dynamical spectral weight transfer to the Hubbard side bands at finite frequencies $\omega \simeq \pm 3$ (in units of the bandwidth). (c) Zoom-in of the central (Kondo) resonance of the impurity spectral function. While undisturbed in the Fermi liquid phase, it splits into a pseudogap in the PG phase, with its height diminishing rapidly as the Mott transition is approached.

Strikingly, this peak shifts very little across the PG regime (see inset), despite a deepening spectral gap and diverging self-energy. Its position lies well below the Mott gap ($\sim 2.5D$), suggesting that it represents an emergent mid-infrared (MIR) excitation intrinsic to the Mott metal characterising the PG. This feature closely mirrors the experimentally observed MIR peak in high- T_c cuprates [64–66], which has been variously attributed to excitations of electrons bound to vacancies [67], or the dressing of holes by spin excitations [68]. Here, its robust energy scale and insensitivity to increasing incoherence imply a universality persisting through the PG, likely rooted in short-range entanglement between deconfined doublons and holons excitations of the Mott metal. This is further evidence

that the PG metal is not simply a degraded Fermi liquid, but a distinct phase whose optical response is driven by emergent degrees of freedom.

1.6.3 Non-local correlations in the pseudogap

In Fig. 1.13(a) and 1.13(b), the spin-flip correlations and mutual information between the impurity spin and conduction bath sites respectively are observed to undergo a crossover within the PG, from a short-ranged behaviour at its onset, to a long-ranged behaviour as the Mott transition approaches. The entanglement is also observed to be multipartite in nature: in Fig. 1.13(c), the quantum Fisher information (QFI) [69] computed for the ground state wavefunction using an operator corresponding to the sum of local spin-flip exchange processes shows a jump at the onset of the PG. Further, the FL is observed to possess bi-partite entanglement while the NFL of the PG phase displays pentapartite entanglement [?, ?].

These striking results imply that the Mott transition observed by us lies beyond the local quantum criticality scenario [70]. Instead, we observe the PG phase to be a novel state of strongly interacting quantum matter emergent from the breakdown of local Kondo screening. This state is described by a quantum critical Fermi surface [71] with NFL Fermi arcs that display increasingly critical behaviour, i.e., dynamics described by non-local quantum fluctuations, and excitations that become truly long-ranged close to the transition. It is tempting to conjecture that the long-range entangled critical state captured in Fig. 1.13(b) reflects the existence of a highly efficient scrambler [72–74], i.e., a state in which entanglement spreads among its constituents at the fastest rate permissible by quantum mechanics.

1.7 Exactly solvable nodal Non-Fermi liquid at Mott Criticality

Very close to the transition, the excitations of the nodal NFL correspond to those of a Hatsugai-Kohmoto model [47, 48]. This insight is obtained from a perturbation-theoretic treatment of the RG fixed point Hamiltonian of the impurity model for $W/J \lesssim (W/J)_{\text{PG}}$, by considering the effects of a small fixed point Kondo scattering probability J^* in the backdrop of a larger bath interaction parameter $|W|$. This yields the HK model [47, 48] as the singular part of the effective Hamiltonian arising from forward scattering processes.

As the bath interaction W is tuned through the L-PG phase, the nodal region is the last to decouple from the impurity. This allows us to write down a simpler Kondo model near the transition, where only the nodal region is hybridising with the impurity spin through Kondo interactions. This is done by retaining only those scattering processes $\mathbf{k}_1 \rightarrow \mathbf{k}_2$ that originate from and end at k -points within a small neighborhood of width $|\mathbf{q}|$ around the four nodal points: $\mathbf{k}_1, \mathbf{k}_2 \in \mathbf{N} + \mathbf{q}$, $|\mathbf{q}| \ll \pi$, where $\mathbf{N}$ can be any one of the four nodal points $\mathbf{N}_1 = (\pi/2, \pi/2)$, $\mathbf{N}_2 = (-\pi/2, \pi/2)$ and $\mathbf{N}_1 + \mathbf{Q}_1$ and $\mathbf{N}_2 + \mathbf{Q}_2$, where $\mathbf{Q}_1 = (-\pi, -\pi)$ and $\mathbf{Q}_2 = (\pi, -\pi)$ are the two nesting vectors. We have already seen, in Subsection 1.5.3, that the pseudogap regime displays a decoupling of the Brillouin zone into sets of points that have no explicit connecting term in the Hamiltonian. Near the transition, the neighbourhoods of $\mathbf{N}_1$ and $\mathbf{N}_1 + \mathbf{Q}_1$ form one such connected set, while the other two nodes $\mathbf{N}_2$ and $\mathbf{N}_2 + \mathbf{Q}_2$ form another connected set, but these two sets are disconnected from each other, in the

sense of a two-channel model. We also assume that the window of $\mathbf{q}$ is small enough so that the fixed point Kondo coupling values $J^*(\mathbf{q}_1, \mathbf{q}_2)$ for the scattering processes involving $\mathbf{q}_1$ and $\mathbf{q}_2$ can be replaced by an average value J^* .

With these considerations, the simplified low-energy model near the transition describing the Kondo scattering processes can be written as

$$\begin{aligned} \tilde{H}_{\text{imp-cbath}} = J^* \frac{1}{2} \sum_{l=1,2} \sum_{\mathbf{q}_1, \mathbf{q}_2} \sum_{\alpha, \beta} \mathbf{S}_d \cdot \boldsymbol{\sigma}_{\alpha\beta} & \left(c_{\mathbf{N}_l+\mathbf{q}_1, \alpha}^\dagger c_{\mathbf{N}_l+\mathbf{q}_2, \beta} + c_{\mathbf{N}_l+\mathbf{Q}_l+\mathbf{q}_1, \alpha}^\dagger c_{\mathbf{N}_l+\mathbf{Q}_l+\mathbf{q}_2, \beta} \right. \\ & \left. - c_{\mathbf{N}_l+\mathbf{Q}_l+\mathbf{q}_1, \alpha}^\dagger c_{\mathbf{N}_l+\mathbf{q}_2, \beta} - c_{\mathbf{N}_l+\mathbf{q}_1, \alpha}^\dagger c_{\mathbf{N}_l+\mathbf{Q}_l+\mathbf{q}_2, \beta} \right) . \end{aligned} \quad (7.1)$$

For scattering processes that involve two opposite nodal points (that is, connected by one of the nesting vectors $\mathbf{Q}_l$, the Kondo coupling is $-J^*$, which explains the negative signs in the third and fourth terms. An explanation of the various dummy indices is in order. The label l can take values 1 or 2, allowing us to consider both the decoupled sets (associated with $\mathbf{N}_1$ and $\mathbf{N}_2$). It also labels the nesting vectors $\mathbf{Q}_l$ associated with the two sets. α and β indicate spin indices, and $\mathbf{q}_1$ and $\mathbf{q}_2$ represent incoming and outgoing momenta in the scattering processes.

Because of the decoupling of the channel $l = 1$ and $l = 2$, we consider only the $l = 1$ channel for the rest of the calculations in this section, and we comment at the end on what the presence of two channels means for the results. In order to simplify the Hamiltonian, we define new fermionic operators

$$\psi_{\mathbf{q}, \sigma} = \frac{1}{\sqrt{2}} (c_{\mathbf{N}_1+\mathbf{q}, \sigma} + c_{\mathbf{N}_1+\mathbf{Q}_1-\mathbf{q}, \sigma}) , \quad \phi_{\mathbf{q}, \sigma} = \frac{1}{\sqrt{2}} (c_{\mathbf{N}_1+\mathbf{q}, \sigma} - c_{\mathbf{N}_1+\mathbf{Q}_1-\mathbf{q}, \sigma}) , \quad (7.2)$$

The operator satisfy fermionic anticommutation relations:

$$\begin{aligned} \left\{ \psi_{\mathbf{q}, \sigma}, \psi_{\mathbf{q}', \sigma'}^\dagger \right\} &= \left[\delta_{\mathbf{q}, \mathbf{q}'} + \frac{1}{2} (\delta_{\mathbf{Q}_1-\mathbf{q}, \mathbf{q}'} + \delta_{\mathbf{q}, \mathbf{Q}_1-\mathbf{q}'}) \right] \delta_{\sigma, \sigma'} \\ &= \delta_{\mathbf{q}, \mathbf{q}'} \delta_{\sigma, \sigma'} , \\ \left\{ \phi_{\mathbf{q}, \sigma}, \phi_{\mathbf{q}', \sigma'}^\dagger \right\} &= \left[\delta_{\mathbf{q}, \mathbf{q}'} - \frac{1}{2} (\delta_{\mathbf{Q}_1-\mathbf{q}, \mathbf{q}'} + \delta_{\mathbf{q}, \mathbf{Q}_1-\mathbf{q}'}) \right] \delta_{\sigma, \sigma'} \\ &= \delta_{\mathbf{q}, \mathbf{q}'} \delta_{\sigma, \sigma'} , \\ \left\{ \psi_{\mathbf{q}, \sigma}, \phi_{\mathbf{q}', \sigma'}^\dagger \right\} &= \left\{ \phi_{\mathbf{q}, \sigma}, \psi_{\mathbf{q}', \sigma'}^\dagger \right\} = 0 , \end{aligned} \quad (7.3)$$

where both $\mathbf{q}$ and $\mathbf{q}'$ are chosen from a small window around the nodes, so their difference is also small and it is therefore guaranteed that $|\mathbf{q} + \mathbf{q}'| \neq |\mathbf{Q}_1|$, rendering both $\delta_{\mathbf{q}, \mathbf{Q}_1-\mathbf{q}'}$ and $\delta_{\mathbf{Q}_1-\mathbf{q}, \mathbf{q}'}$ zero. For convenience, we define new number operators for the sum and relative degrees of freedom:

$$s_{\mathbf{q}, \sigma} = \psi_{\mathbf{q}, \sigma}^\dagger \psi_{\mathbf{q}, \sigma} , \quad r_{\mathbf{q}, \sigma} = \phi_{\mathbf{q}, \sigma}^\dagger \phi_{\mathbf{q}, \sigma} . \quad (7.4)$$

We then have the following useful relation between the corresponding number operators $n = c^\dagger c$:

$$n_{\mathbf{N}_1+\mathbf{q}, \sigma} + n_{\mathbf{N}_1+\mathbf{Q}_1-\mathbf{q}, \sigma} = s_{\mathbf{q}, \sigma} + r_{\mathbf{q}, \sigma} . \quad (7.5)$$

We find that the sum degrees of freedom $s_{\mathbf{q},\sigma}$ do not enter the Kondo scattering Hamiltonian $\tilde{H}_{\text{imp-cbath}}$ (for the single channel $l = 1$), and it is expressed purely in terms of the relative degrees of freedom:

$$\tilde{H}_{\text{imp-cbath}} = J^* \sum_{\mathbf{q}_1, \mathbf{q}_2, \alpha, \beta} \mathbf{S}_d \cdot \sigma_{\alpha\beta} \phi_{\mathbf{q}_1\alpha}^\dagger \phi_{\mathbf{q}_2\beta} . \quad (7.6)$$

We now proceed to rewrite the other parts of the Hamiltonian in terms of these new degrees of freedom. The kinetic energy can be written as

$$\begin{aligned} \tilde{H}_{\text{cbath}} &= \sum_{\mathbf{q}_1, \mathbf{q}_2, \sigma} \varepsilon_{\mathbf{N}_1 + \mathbf{q}} (n_{\mathbf{N}_1 + \mathbf{q}, \sigma} + n_{\mathbf{N}_1 + \mathbf{Q}_1 - \mathbf{q}, \sigma}) \\ &= \sum_{\mathbf{q}, \sigma} \varepsilon_{\mathbf{N}_1 + \mathbf{q}} (s_{\mathbf{q}, \sigma} + r_{\mathbf{q}, \sigma}) , \end{aligned} \quad (7.7)$$

Given that the Kondo coupling acts only on the subspace of the ϕ fields, we consider a simplified form of the bath interaction (for the channel $l = 1$), taking into account scattering processes restricted to the ϕ fields:

$$\tilde{H}_{\text{cbath-int}} = -\frac{1}{2N} \sum'_{\mathbf{q}_1, \mathbf{q}_2, \sigma} \frac{W}{2} c_{\mathbf{q}_1, \sigma}^\dagger c_{\mathbf{q}_2, \sigma} + \frac{1}{4N^2} \sum'_{\{\mathbf{q}_i\}} W c_{\mathbf{q}_1, \uparrow}^\dagger c_{\mathbf{q}_2, \uparrow} c_{\mathbf{q}_3, \downarrow}^\dagger c_{\mathbf{q}_4, \downarrow} , \quad (7.8)$$

where the primed summations indicates each $\{\mathbf{q}_i\}$ in both terms sum over the neighbourhoods of N_1 and $N_1 + \mathbf{Q}_1$, and each Fourier transform contributes a factor of $1/\sqrt{2N}$ (if N is the number of points on a single arm of the Fermi surface, the total number of points in $l = 1$ is $2N$). Any single primed summation can be written as

$$\sum'_q c_{q, \sigma} = \sum_q (c_{N_1 + q, \sigma} - c_{N_1 + \mathbf{Q}_1 - q, \sigma}) , \quad (7.9)$$

where the unprimed summation only sums over the neighborhood of N_1 . The change in sign arises because every transfer of momentum from N_1 to $N_1 + \mathbf{Q}_1$ introduces a phase of π within the k -dependence of W . Noting that $c_{N_1 + q, \sigma} - c_{N_1 + \mathbf{Q}_1 - q, \sigma} = \sqrt{2} \phi_q^\dagger$, we get

$$\tilde{H}_{\text{cbath-int}} = -\frac{W}{2} \sum_{\mathbf{q}_1, \mathbf{q}_2, \sigma} \phi_{\mathbf{q}_1, \sigma}^\dagger \phi_{\mathbf{q}_2, \sigma} + \frac{W}{N^2} \sum_{\{\mathbf{q}_i\}} \phi_{\mathbf{q}_1, \uparrow}^\dagger \phi_{\mathbf{q}_2, \uparrow} \phi_{\mathbf{q}_3, \downarrow}^\dagger \phi_{\mathbf{q}_4, \downarrow} . \quad (7.10)$$

We now argue that that the ground state of $\tilde{H}_{\text{cbath-int}}$ is the state

$$|L\rangle = \frac{1}{N} \sum_{\mathbf{q}_1 \mathbf{q}_2} \phi_{\mathbf{q}_1, \uparrow}^\dagger \phi_{\mathbf{q}_2, \downarrow}^\dagger |0\rangle . \quad (7.11)$$

This can be seen by defining new “local” degrees of freedom $\tilde{\phi}(r)_\sigma = \frac{1}{\sqrt{N}} \sum_k e^{i\vec{k} \cdot \vec{r}} \phi_{k, \sigma}$. In terms of them, the Hamiltonian takes the form

$$\tilde{H}_{\text{cbath-int}} = -\frac{W}{2} \sum_\sigma \phi^\dagger(0)_\sigma \phi(0)_\sigma + W \phi^\dagger(0)_\uparrow \phi(0)_\uparrow \phi^\dagger(0)_\downarrow \phi(0)_\downarrow . \quad (7.12)$$

This becomes a four-level problem associated with the two number operators $\phi^\dagger(0)_\sigma \phi(0)_\sigma$ that commute with the Hamiltonian and since $W < 0$, the ground state is doubly degenerate between the completely unoccupied and the doubly occupied states. The state in eq. 7.11 is the doubly occupied state:

$$|L\rangle = \phi^\dagger(0)_\uparrow \phi^\dagger(0)_\downarrow |0\rangle. \quad (7.13)$$

This can also be verified by working with the momentum-space ϕ operators.

Since both $\tilde{H}_{\text{imp-cbath}}$ and $\tilde{H}_{\text{cbath-int}}$ commute with $s_{\mathbf{q},\sigma}$, we replace the local objects $\sum_{\mathbf{q}} s_{\mathbf{q},\sigma}$ in $\tilde{H}_{\text{cbath-int}}$ with $\frac{1}{2}$ since the bath is always at half-filling. This results in a simplified Hamiltonian for just the $\phi_{\mathbf{q},\sigma}$ degrees of freedom:

$$\tilde{H} = -\frac{1}{2}W \sum_{\mathbf{q},\sigma} r_{\mathbf{q},\sigma} + W \sum_{\mathbf{q}_1, \mathbf{q}_2} r_{\mathbf{q}_1, \uparrow} r_{\mathbf{q}_2, \downarrow} + \sum_{\mathbf{q}_1, \mathbf{q}_2, \alpha, \beta} J^* \mathbf{S}_d \cdot \boldsymbol{\sigma}_{\alpha\beta} \phi_{\mathbf{q}_1\alpha}^\dagger \phi_{\mathbf{q}_2\beta} \quad (7.14)$$

For bath interaction strength close to the critical value ($W \lesssim W_c$), the fixed point coupling value J^* is much smaller than W . In order to obtain the gapless excitations of the system arising from the presence of the impurity site, we integrate out the impurity dynamics via a Schrieffer-Wolff transformation. The perturbation term $\mathcal{V}$ then consists of Hamiltonian terms that modify the impurity configuration,

$$\mathcal{V} = \sum_{\mathbf{q}_1, \mathbf{q}_2} J^* S_d^+ \phi_{\mathbf{q}_1\downarrow}^\dagger \phi_{\mathbf{q}_2\uparrow} + \text{h.c.}, \quad (7.15)$$

while the “non-interacting” Hamiltonian is

$$H_D = -\frac{1}{2}W \sum_{\mathbf{q},\sigma} r_{\mathbf{q},\sigma} + W \sum_{\mathbf{q}_1, \mathbf{q}_2} r_{\mathbf{q}_1, \uparrow} r_{\mathbf{q}_2, \downarrow}. \quad (7.16)$$

The renormalisation in the dynamics of $b_{\mathbf{k},\sigma}^\dagger$ are given by

$$\Delta\tilde{H} = \mathcal{P}_L \mathcal{V} \mathcal{P}_H \frac{1}{E_L - H_D} \mathcal{V} \mathcal{P}_L, \quad (7.17)$$

where E_L is the energy of the ground state of H_D , $-(E_L - H_D)$ is the excitation energy of the state generated upon applying the perturbation $\mathcal{V}$, and $\mathcal{P}_{L(H)}$ projects onto the low(high)-energy subspace. For the present Hamiltonian, the low-energy state (associated with E_L) is the one that minimises the bath interaction term $\tilde{H}_{\text{cbath-int}}$; since the coupling W is negative, the minimal configuration is one in which the local operators $\sum_{\mathbf{q}} r_{\mathbf{q},\sigma}$ (for $\sigma = \pm 1$) are unity. This leads to the following low-energy state:

$$|L\rangle = \frac{1}{N} \sum_{\mathbf{q}_1, \mathbf{q}_2} \phi_{\mathbf{q}_1, \uparrow}^\dagger \phi_{\mathbf{q}_2, \downarrow}^\dagger |0\rangle \quad (7.18)$$

where $|0\rangle$ is the $r_{\mathbf{q}\uparrow} = r_{\mathbf{q}\downarrow} = 0$ state. The state $|L\rangle$ is at zero energy. High-energy states are obtained by applying, on the state $|L\rangle$, the excitation operator $\phi_{\mathbf{q}_1, \sigma}^\dagger \phi_{\mathbf{q}_2, \bar{\sigma}}$ or its hermitian conjugate. The operator converts a double and a hole to spin states, incurring a total excitation cost of $-W = |W|$.

We now calculate the various terms in $\Delta\tilde{H}$:

$$\begin{aligned}
 \mathcal{P}_L \mathcal{V} \mathcal{P}_H \frac{1}{E_L - H_D} \mathcal{V} \mathcal{P}_L &= \left(\frac{1}{2} + S_d^z \right) \sum_{\mathbf{q}_1, \mathbf{q}_2} \frac{J^{*2}}{-4W} \phi_{\mathbf{q}_1, \downarrow}^\dagger \phi_{\mathbf{q}_2, \uparrow} \left(\phi_{\mathbf{q}_2, \uparrow}^\dagger \phi_{\mathbf{q}_1, \downarrow} + (1 - \delta_{\mathbf{q}_1, \mathbf{q}_2}) \phi_{\mathbf{q}_1, \uparrow}^\dagger \phi_{\mathbf{q}_2, \downarrow} \right) \\
 &= - \sum_{\mathbf{q}_1, \mathbf{q}_2} \frac{J^{*2}}{4W} \left[r_{\mathbf{q}_1 \downarrow} (1 - r_{\mathbf{q}_2 \uparrow}) - (1 - \delta_{\mathbf{q}_1, \mathbf{q}_2}) \phi_{\mathbf{q}_1, \uparrow}^\dagger \phi_{\mathbf{q}_1, \downarrow}^\dagger \phi_{\mathbf{q}_2, \downarrow} \phi_{\mathbf{q}_2, \uparrow} \right] \\
 \mathcal{P}_L \mathcal{V} \mathcal{P}_H \frac{1}{E_L - H_D} \mathcal{V} \mathcal{P}_L &= \sum_{\mathbf{q}_1, \mathbf{q}_2} \frac{J^{*2}}{-4W} \left[r_{\mathbf{q}_1 \uparrow} (1 - r_{\mathbf{q}_2 \downarrow}) - (1 - \delta_{\mathbf{q}_1, \mathbf{q}_2}) \phi_{\mathbf{q}_1, \downarrow}^\dagger \phi_{\mathbf{q}_1, \uparrow}^\dagger \phi_{\mathbf{q}_2, \uparrow} \phi_{\mathbf{q}_2, \downarrow} \right]
 \end{aligned} \tag{7.19}$$

In both terms, we set $S_d^z = 0$ owing to local SU(2) symmetry on the impurity site. In the same way, we treat the effects of the dispersion term (eq. 7.7) to second order:

$$\begin{aligned}
 \mathcal{P}_L \mathcal{V} \mathcal{P}_H \frac{1}{E_L - H_D} \mathcal{V} \mathcal{P}_L &= \sum_{\mathbf{q}, \sigma} \frac{\varepsilon_{\mathbf{N}_1 + \mathbf{q}}^2}{-W} \left[\psi_{\mathbf{q}, \sigma}^\dagger \phi_{\mathbf{q}, \sigma} \phi_{\mathbf{q}, \sigma}^\dagger \psi_{\mathbf{q}, \sigma} + \phi_{\mathbf{q}, \sigma}^\dagger \psi_{\mathbf{q}, \sigma} \psi_{\mathbf{q}, \sigma}^\dagger \phi_{\mathbf{q}, \sigma} \right] \\
 &= \sum_{\mathbf{q}, \sigma} \frac{\varepsilon_{\mathbf{N}_1 + \mathbf{q}}^2}{-W} \left[s_{\mathbf{q}, \sigma} (1 - r_{\mathbf{q}, \sigma}) + r_{\mathbf{q}, \sigma} (1 - s_{\mathbf{q}, \sigma}) \right].
 \end{aligned} \tag{7.20}$$

At the values of $\mathbf{q}$ for which $\varepsilon_{\mathbf{N}_1 + \mathbf{q}} > 0$, we can set the number operator $s_{\mathbf{q}, \sigma} = 0$ (these positive energy modes are unoccupied in the ground state), while for $\varepsilon_{\mathbf{N}_1 + \mathbf{q}} < 0$, we have $s_{\mathbf{q}, \sigma} = 1$:

$$\mathcal{P}_L \mathcal{V} \mathcal{P}_H \frac{1}{E_L - H_D} \mathcal{V} \mathcal{P}_L = \sum_{\mathbf{q} (\varepsilon_{\mathbf{N}_1 + \mathbf{q}} > 0), \sigma} \frac{\varepsilon_{\mathbf{N}_1 + \mathbf{q}}^2}{-W} r_{\mathbf{q}, \sigma} + \sum_{\mathbf{q} (\varepsilon_{\mathbf{N}_1 + \mathbf{q}} < 0), \sigma} \frac{\varepsilon_{\mathbf{N}_1 + \mathbf{q}}^2}{-W} (1 - r_{\mathbf{q}, \sigma}). \tag{7.21}$$

The complete second-order renormalised Hamiltonian is

$$\Delta\tilde{H} = \sum_{\mathbf{q}, \sigma} \epsilon_{\mathbf{q}} r_{\mathbf{q}, \sigma} + \mathcal{U} \sum_{\mathbf{q}, \sigma} r_{\mathbf{q}, \sigma} r_{\mathbf{q}, \bar{\sigma}} + \mathcal{U} \sum_{\mathbf{q}_1 \neq \mathbf{q}_2, \sigma} \left[r_{\mathbf{q}_1, \sigma} r_{\mathbf{q}_2, \bar{\sigma}} + \phi_{\mathbf{q}_1, \bar{\sigma}}^\dagger \phi_{\mathbf{q}_1, \sigma}^\dagger \phi_{\mathbf{q}_2, \sigma} \phi_{\mathbf{q}_2, \bar{\sigma}} \right]. \tag{7.22}$$

where $\epsilon_{\mathbf{q}} = \text{sign}(\varepsilon_{\mathbf{N}_1 + \mathbf{q}}) \frac{\varepsilon_{\mathbf{N}_1 + \mathbf{q}}^2}{-W}$ and $\mathcal{U} = \frac{J^{*2}}{4W}$.

The $\mathbf{q}_1 = \mathbf{q}_2$ component of the Hamiltonian shows the emergence of the exactly solvable Hatsugai-Kohmoto model [47, 48] at the critical point. The correlation term $\mathcal{U} \sum_{\mathbf{q}, \sigma} r_{\mathbf{q}, \sigma} r_{\mathbf{q}, \bar{\sigma}}$ leads to a transfer of spectral weight across the Fermi surface and separates the available states into three classes:

$$\langle n_{\mathbf{q}} \rangle = \begin{cases} 2, & \epsilon_{\mathbf{q}} < -|\mathcal{U}/2|, \\ 1, & |\mathcal{U}/2| > \epsilon_{\mathbf{q}} > -|\mathcal{U}/2|, \\ 0, & |\mathcal{U}/2| < \epsilon_{\mathbf{q}}. \end{cases} \tag{7.23}$$

In contrast to a Fermi liquid, there is a highly degenerate single-occupied region in the middle, and this gives rise to non-Fermi liquid excitations.

We now discuss the effects of the $\mathbf{q}_1 \neq \mathbf{q}_2$ component. The first term partially lifts the degeneracy of the central singly-occupied region and allows only zero magnetisation configurations. The second

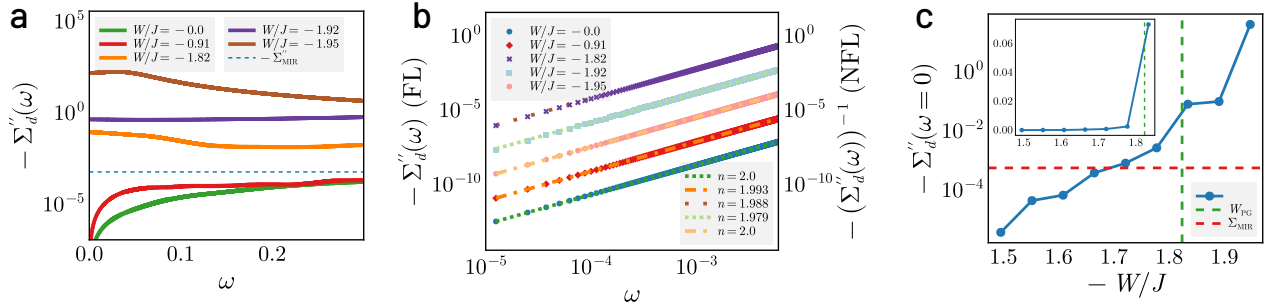


Figure 1.11: (a) Imaginary part of impurity self-energy $\Sigma''(\omega)$ as a function of frequency ω for Fermi liquid (FL, $|W/J| < 1.79$) and pseudogapped phases (NFL, $|W/J| \geq 1.79$). $\Sigma''(\omega)$ falls to zero as $\omega \rightarrow 0$ for the FL, while it attains a peak in the pseudogap. All $\Sigma''(\omega)$ for the NFL are observed to lie above the Mott-Ioffe-Regel (MIR) bound (dashed blue line), while those for the FL lie below. (b) Scaling of $\Sigma''(\omega)$ with frequency for Fermi liquid and pseudogapped phases. The FL self-energy fits to $\Sigma'' \sim \omega^\alpha$ with $\alpha \approx 2$, vanishing as $\omega \rightarrow 0$, while the NFL self-energy grows as $\Sigma'' \approx a - \omega^\beta$ for small ω , with $\beta \approx 2$. Remarkably, the NFL exponent remains mostly unchanged through the entirety of the PG phase. (c) Variation of the zero-frequency imaginary self-energy $-\Sigma''(\omega = 0)$ with ω in the FL and pseudogap phases (entry into the PG is marked by the vertical dashed line). The inset shows the same but in linear scale, in order to display the dramatic rise (by almost 30 times) on entering the PG.

term creates gapped excitations involving the regions of zero and double occupancy; these represent subdominant pairing fluctuations of the nodal non-Fermi liquid. We recall that similar fluctuations across the Fermi surface were observed in our calculations of charge correlations in Fig. 1.9 (a), and are consistent with pairing fluctuations observed from quantum Monte Carlo simulations and found to give rise to quantum-critical behavior and PG physics [28].

Consequently, the resulting NFL metal of the lattice model involves long-lived excitations of multiple $\mathbf{k}$ -states, and manifest in the form of a divergent one-particle self-energy at the non-interacting FS [75]: $\Sigma_{\mathbf{q}}(\omega) = -\mathcal{U}^2/4(\omega - \epsilon_{\mathbf{q}})$, such that $\Sigma_{\epsilon_{\mathbf{q}}=0}(\omega \rightarrow 0) \rightarrow \infty$. This zero-frequency self-energy pole presages the transition into a Mott insulating phase, where it marks a hard gap in the spectral function for charge excitations. This is consistent with our findings for the lattice (Fig. 1.9(c)) and impurity self-energies (Fig. 1.11(c)). The nodal Mott metal thus comprises of a Greens function zero at the Fermi energy together with an anisotropic massless Dirac dispersion, leading to a non-zero Chern number [76–78]. This topological index survives into the Mott insulator.

For small but non-zero values of $\omega - \epsilon_{\mathbf{k}}$, we obtain a quasiparticle residue that vanishes with ω , $Z_{\text{imp}} \sim \omega^2/\mathcal{U}^2$. The scattering rate of this singular NFL possesses a sharp peak at the FS ($\omega = \epsilon_{\mathbf{k}_F}$):

$$\Gamma \sim \mathcal{U}^2 \delta(\omega - \epsilon_{\mathbf{k}}), \quad (7.24)$$

consistent with the sharply peaked Lorentzian $\Sigma''_{\mathbf{k}}(\omega) \sim \omega^{-2}$ captured in Fig. 1.11(a). This quantum critical NFL metal is an example of a strongly coupled scale-invariant form of quantum matter. The exact solution for eq.(?) reveals the presence of low-energy excitations comprised of holons and doublons [47]. These features point to the nodal NFL as a long-ranged and multipartite entangled, strongly interacting scale-invariant state of quantum matter [2, 79–81] that are completely

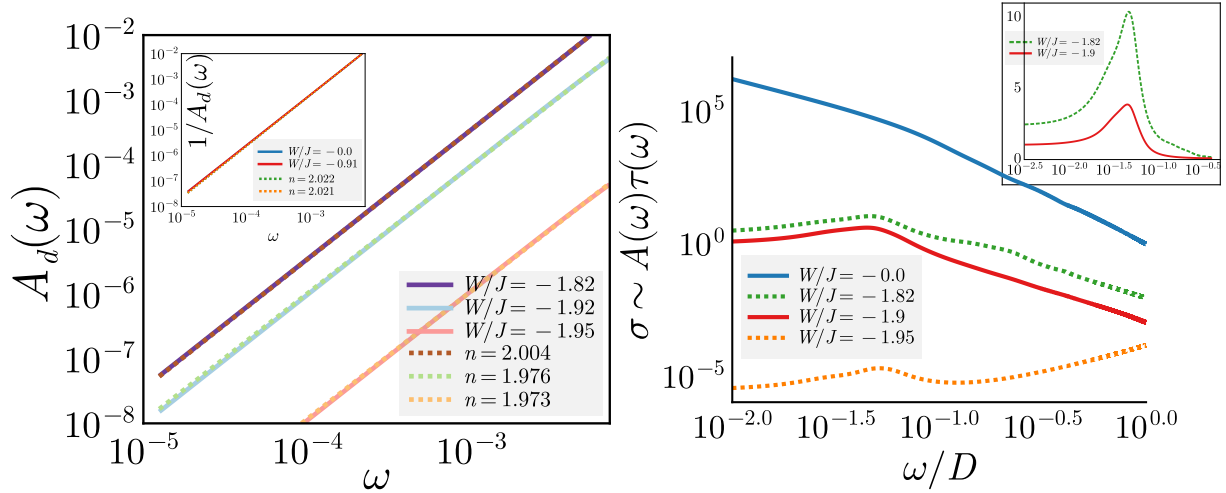


Figure 1.12: Left: Impurity spectral function $A_d(\omega)$ in the local Fermi liquid (inset) and pseudogap (main). The PG spectral function is fit to a power law (with the $\omega = 0$ contribution subtracted, see Fig. 1.11(c) for the full spectral function). The exponent of the power law comes out to be $n \approx 2$ throughout the PG. The local FL spectral function (inset) is fit to a lorentzian (by fitting the inverse to a power law). Right: Optical conductivity $\sigma(\omega)$ for excitations of the impurity model, approximated through the product of the carrier density (taken from the density of states $A(\omega)$) and the lifetime $\tau(\omega)$ (obtained from the inverse scattering rate $-1/\Sigma''(\omega)$, where Σ'' is the imaginary part of the self-energy). The Fermi liquid (blue) shows a sharp Drude peak at $\omega = 0$ due to the presence of long-lived quasiparticles. The PG (green, red, orange) shows a shifted Drude peak at a non-zero frequency which is reminiscent of the mid-infrared (MIR) peak seen experimentally in the cuprates [64–66]. Inset shows the robustness of the position of the MIR peak through the PG, indicating it as a universal feature of the Mott metal.

disconnected from the quasiparticles of the FL. Following the arguments laid out in [5], at finite temperatures, such a scale-invariant highly entangled non-Fermi liquid is likely associated with Planckian dissipation (i.e., $\Sigma''(T) \sim k_B T$) and a resistivity that varies linearly with temperature ($\rho \sim T$) [?, 82, 83]. This sits well with the possibility that this state is also a highly efficient scrambler [72–74]. Additionally, as mentioned above, we observe that the nodal metal possesses pairing fluctuations that can become dominant upon doping [75].

1.8 Heisenberg Model as a low-energy description of the tiled Mott insulator

For simplicity, we consider two impurity sites labelled 1 and 2 associated with two nearest-neighbour auxiliary models. The ground state subspace is four-fold degenerate:

$$|\Psi_L\rangle = \{|\sigma_1, \sigma_2\rangle\}, \quad \sigma_i = \pm 1, \quad (8.1)$$

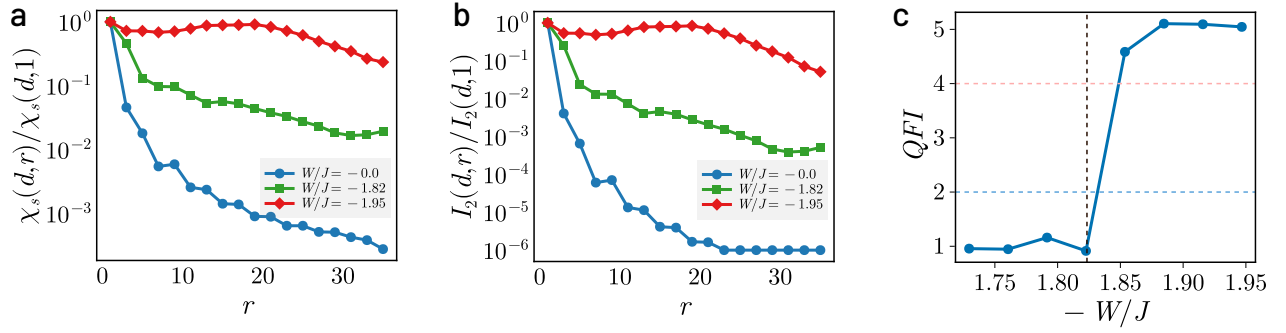


Figure 1.13: (a) Spin correlation $\langle \mathbf{S}_d \cdot \mathbf{S}_r \rangle$ and (b) mutual information $I_2(d, r)$ between the impurity spin and conduction bath local spin density as a function of the distance $\mathbf{r}$ between them, normalised against the value at $r = 1$. Both decay very quickly in the FL phase (blue), but show long-ranged behaviour in the NFL phase (green and red), extending to the edges of the system at the critical point (red). (c) Evolution of the Quantum Fisher Information F_Q for a nearest-neighbour spin-flip operator $O = \sum_{i \in \text{odd}} (S_i^+ S_{i+1}^- + \text{h.c.})$ through the first Lifshitz transition and the pseudogap. The vertical dashed line marks the onset of the PG. To the left of it, the QFI in the Fermi liquid phase shows at most bipartite entanglement ($F_Q < 2$ (below blue dashed line)), while the PG shows the presence of multipartite entanglement upto 5 parties ($F_Q > 4$ (red dashed line)).

where σ_i is the spin state of site i . This ground state is derived from the following “zeroth order” Hamiltonian that emerges in the local moment phase of the auxiliary models when all scattering processes between the impurity and conduction bath are RG-irrelevant:

$$H_0 = -\frac{U}{2} \sum_{i=1,2} (n_{i\uparrow} - n_{i\downarrow})^2 ; \quad (8.2)$$

the local correlation on the impurity site becomes the largest scale in the problem in this phase and pushes the $|n_i = 2\rangle$ and $|n_i = 0\rangle$ states to high energies. This then defines the high-energy subspace for our calculation:

$$|\Psi_H\rangle = |C_1, C_2\rangle , \quad (8.3)$$

where C_i can take values 0 or 2, indicating that the state i is either empty or full, respectively. Both the double and hole states exist at a charge gap of the order of $U/2$ above the low-energy singly-occupied subspace defined by the states $|\Psi_L\rangle$.

In order to allow virtual fluctuations that can lift the large ground state degeneracy and lower the energy, we consider (perturbatively) the effects of an irrelevant single-particle hybridisation that connects the nearest-neighbour sites. This perturbation Hamiltonian is therefore of the form

$$H_t = \sum_{\omega} V(\omega) \mathcal{P}(\omega) \sum_{\sigma} \left(c_{1\sigma}^{\dagger} c_{2\sigma} + \text{h.c.} \right) , \quad (8.4)$$

where $V(\omega)$ only acts on states at the energy scale ω ; the renormalisation of V is encoded in the fact that $V(\omega)$ is largest for the excited states and vanishes at low-energies: $V(\omega \rightarrow 0) = 0$.

In order to obtain a low-energy effective Hamiltonian for the impurity sites arising from this hybridisation, we integrate out H_t via a Schrieffer-Wolff transformation. This leads to the following second-order Hamiltonian:

$$H_{\text{eff}} = \mathcal{P}_L H_t G \mathcal{H}_i \mathcal{P}_L . \quad (8.5)$$

The operator $\mathcal{P}_L$ projects onto the low-energy subspace $|\Psi_L\rangle$ - this ensures that we remain in the low-energy subspace at the beginning and at the end of the total process. The Greens function $G = (E_L - H_0)^{-1}$ incorporates the excitation energy to go from the low-energy subspace $|\Psi\rangle_L$ (of energy E_L) to the excited subspace $|\Psi\rangle_H$ of energy $E_L + U/2$. Substituting the form of the perturbation Hamiltonian and the excitation energy into the above expression gives

$$H_{\text{eff}} = \frac{V_H^2}{-U/2} \sum_{\sigma, \sigma'} \left[c_{1\sigma}^\dagger c_{2\sigma} c_{2\sigma'}^\dagger c_{1\sigma'} + c_{2\sigma}^\dagger c_{1\sigma} c_{1\sigma'}^\dagger c_{2\sigma'} \right] . \quad (8.6)$$

where $V_H \equiv V(\omega \rightarrow U/2)$ is the impurity-bath hybridisation at energy scales of the order of the Mott gap, in the sense of an RG flow. Terms with consecutive creation or annihilation operators on the same site are prohibited because each site is singly-occupied in the ground state. It is now easy to cast this Hamiltonian into a more recognizable form. For $\sigma' = \sigma$, we get

$$\sum_{\sigma} \delta_{\sigma, \sigma'} c_{1\sigma}^\dagger c_{2\sigma} c_{2\sigma'}^\dagger c_{1\sigma'} = \sum_{\sigma} (n_{1\sigma} - n_{1\sigma} n_{2\sigma}) , \quad (8.7)$$

while $\sigma = -\sigma' = \pm 1$ gives

$$\sum_{\sigma} \delta_{\sigma, -\sigma'} c_{1\sigma}^\dagger c_{2\sigma} c_{2\sigma'}^\dagger c_{1\sigma'} = - (S_1^+ S_2^- + \text{h.c.}) . \quad (8.8)$$

For the latter expression, we introduced the local spin-flip operators $S_i^\pm$. The expression above it can also be cast into spin variables, using the equations

$$\begin{aligned} \frac{1}{2} \sum_{\sigma} n_{i\sigma} &= \frac{1}{2}, \\ \frac{1}{2} \sum_{\sigma} \sigma n_{i\sigma} &= S_i^z, \end{aligned} \quad (8.9)$$

where the first equation is simply the condition of half-filling at each site, and the second equation is the definition of the local spin operator in z -direction. Adding and subtracting the equations gives $n_{i\sigma} = \frac{1}{2} + \sigma S_i^z$.

Substituting everything back into eq. 8.6 and dropping constant terms gives

$$H_{\text{eff}} = 2 \frac{V_H^2}{U/2} (2S_1^z S_2^z + S_1^+ S_2^- + S_1^- S_2^+) = J_{\text{eff}} \mathbf{S}_1 \cdot \mathbf{S}_2 , \quad (8.10)$$

where the effective antiferromagnetic Heisenberg coupling is $J_{\text{eff}} = \frac{8V_H^2}{U}$.

1.9 Pseudogap as a strongly coupled phase of quantum matter

We now unveil an organising principle that leads to the remarkable properties observed above for the strongly interacting NFL of the PG phase. The existence of a sharp connected FS at $T = 0$ can be understood as the existence of a topologically protected manifold of gapless chiral excitations in $\mathbf{k}$ -space at the FS [50]. The FS is characterised by a topological index corresponding to an anomaly in the quantum many-body theory for electrons, and can be understood as a generalised symmetry of such a system [51, 84]. A theorem by Luttinger and Ward [85] shows that a count of the physical charge (known as Luttinger’s volume) is identical to the topological index (a so-called homotopy charge known as Luttinger’s count) even in the presence of electronic interactions that do not disturb the FS. We will now argue that the emergence of antinodal Luttinger surfaces involve a disconnection of the FS (into Fermi arcs) and that, by following La Nave et al. [51], the accompanying change in its topological properties leads to the existence of gapless NFL excitations that are non-local in nature.

The antinodal Luttinger surfaces arise from the splitting of double poles of the single-particle Greens function on the FS into poles lying on opposite complex half-planes, together with zeros that are pinned at the FS. These changes in the analytic structure of the single-particle Greens function have important consequences. First, the emergent zeros break a $\mathbb{Z}_2$ symmetry of the FS [41, 42, 51]. This symmetry is guaranteed within FL theory because the presence of quasiparticles vanishingly close to the FS allows the interchange of spin and charge degrees of freedom, promoting the separate $SU(2)$ symmetries of spin and charge to the larger symmetry of $O(4) = SU(2) \times SU(2) \times \mathbb{Z}_2$ [41]. The insertion of zeros of the Greens function at the Fermi surface in the PG, and the associated splitting of the Greens function, breaks this $\mathbb{Z}_2$ symmetry by placing the pole for the spin excitation in one half of the complex plane while placing that for the charge excitation in the other half [42, 86].

Second, they signal a divergent electronic self-energy as a function of the wavevector $\mathbf{k}$, render ill-behaved the Luttinger-Ward functional of the interacting electronic problem, and violate the generalised symmetry encoded within it. The changes in the pole structure change the Luttinger count topological invariant, while the zeros give rise to an additional topological contribution (linked to the Adler-Bell-Jackiw-type chiral anomaly) [49, 78, 87, 88]. As a consequence of the half-filled particle-hole symmetric nature of the system at hand, Luttinger’s volume is preserved upon taking into account topological contributions from both the Luttinger count *and* the zeros [40]. Importantly, La Nave et al. [51] argue, following recent developments in understanding generalised symmetries [89], that the additional anomaly arising from the Luttinger surfaces guarantees the existence of gapless NFL excitations that are non-local in nature.

Thus, the drastic change in nature of the real-space excitations - from locally well-defined Landau quasiparticles of the FL to the increasingly nonlocal excitations of the NFL Fermi arcs in the PG phase - appears to be dictated by a topological principle and, therefore, robust under renormalisation. This signals the NFL Fermi arcs of the PG as an emergent phase of strongly interacting quantum matter - which we dub the *Mott metal* - that is the parent metal of the MI. Such an identification of the pseudogap as a distinct phase is backed up by angle-resolved photoemission spectroscopy (ARPES) experiments [8]; the anomalous momentum, temperature and doping dependence of the gap function observed in the cuprates suggests that the pseudogap maybe described by an order parameter distinct from that of the superconductor [90, 91]. The same topological principle connects the nonlocal unparticle-like gapless excitations [2, 79–81] of the scale-invariant nodal NFL of the

HK model observed precisely at the Mott critical point to those of the rest of the PG phase, e.g., a universal scaling of $\Sigma''_{\mathbf{k}}(\omega) \sim (a + \omega^2)^{-1}$ (Fig. 1.11(b)) and local density of states $A_d(\omega) \sim \omega^2$ (Fig. 1.12) of the NFL throughout the PG phase, such that the value of $\Sigma''_{\mathbf{k}}(\omega = 0)$ value continues to grow upon violating the MIR bound (Fig. 1.11(c)).

1.10 Conclusions

Our analysis reveals that the Mott transition proceeds via a continuous evolution through a PG regime characterized by a singular NFL metal - the Mott metal - with deconfined holon–doublon excitations confined to nodal Fermi arcs, and a scaling of it’s spectral features with universal power-law exponents. As the system approaches criticality, this metallic phase exhibits increasingly non-local correlations and a divergent self-energy, signalling the breakdown of Landau quasiparticles, formation of local moments and the onset of long-range quantum entanglement. Anchored in two-channel Kondo dynamics at intermediate scales and governed by Hatsugai–Kohmoto physics near the critical endpoint, the Mott metal provides a unified framework for understanding anomalous metallicity in strongly correlated systems. It’s fate under finite doping presents a compelling direction for future investigation. More complicated models can be studied by suitably extending the tiling approach and choosing an appropriate auxiliary model. Electron-phonon coupling can be taken into account by studying an Anderson-Holstein model that couples the impurity site to local lattice modes [92]. The effects of disorder can be studied by averaging quantities over various disorder configurations of the conduction bath. The versatility of the unitary RG solver and the impurity model-based construction makes it convenient to explore a large variety of lattice models.

1.11 Appendix: Effective action for the Hubbard-Heisenberg model and the eSIAM

In order to better understand the similarities and differences between the tiling approach outlined here and the approach adopted in dynamical mean-field approximations, we compute the effective action for the local impurity as well as the impurity-zeroth site system, in both the auxiliary and bulk models. This will be done in the limit of infinite coordination number, in order to create a tractable effective theory.

1.11.1 Local theory for the Hubbard-Heisenberg model

We will first work with the Hubbard-Heisenberg model. Let us recall that within the eSIAM, most of the dynamics is governed by the physics of the impurity site and the bath site coupled to the impurity. Accordingly, we will first obtain an effective action within the Hubbard-Heisenberg model for the small system consisting of a local site and all its nearest-neighbours.

We choose a certain local site that we call 0, whose nearest-neighbours will be denoted as $\{\bar{0}\}$. The action for the full Hubbard-Heisenberg model can be formally separated into three parts:

$$S_{\text{H-H}} = S_{0,\bar{0}} + S^{(0,\bar{0})} + S_{\text{int}} , \quad (11.1)$$

where $S_{0,\bar{0}}$ represents the part of the action that involves only the sites 0 and $\{\bar{0}\}$, $S^{(0,\bar{0})}$ represents the part that has all the sites apart from 0 and $\{\bar{0}\}$, while S_{int} contains all terms that connect these two parts. These three parts have the forms

$$S_{0,\bar{0}} = \sum_{i=0,\bar{0}} \sum_{\sigma} \int_0^{\beta} d\tau c_{i\sigma}^{\dagger}(\tau) \partial_{\tau} c_{i\sigma}(\tau) - t \sum_{\bar{0},\sigma} \int_0^{\beta} d\tau \left[c_{0\sigma}^{\dagger}(\tau) c_{\bar{0}\sigma}(\tau) + \text{h.c.} \right] \\ - \frac{U}{2} \sum_{i=0,\bar{0}} \int_0^{\beta} d\tau (n_{i\uparrow}(\tau) - n_{i\downarrow}(\tau))^2 + J \sum_{\bar{0}} \int_0^{\beta} d\tau \vec{S}_0(\tau) \cdot \vec{S}_{\bar{0}}(\tau), \quad (11.2)$$

$$S^{(0,\bar{0})} = \sum_{j \neq (0,\bar{0})} \sum_{\sigma} \int_0^{\beta} d\tau c_{j\sigma}^{\dagger}(\tau) \partial_{\tau} c_{j\sigma}(\tau) - t \sum_{\langle j,l \rangle \neq (0,\bar{0}),\sigma} \int_0^{\beta} d\tau \left[c_{j\sigma}^{\dagger}(\tau) c_{l\sigma}(\tau) + \text{h.c.} \right] \\ - \frac{U}{2} \sum_{j \neq (0,\bar{0})} \int_0^{\beta} d\tau (n_{j\uparrow}(\tau) - n_{j\downarrow}(\tau))^2 + J \sum_{\langle j,l \rangle \neq (0,\bar{0})} \int_0^{\beta} d\tau \vec{S}_l(\tau) \cdot \vec{S}_j(\tau), \quad (11.3)$$

$$S_{\text{int}} = -t \sum_{i \in \bar{0}} \sum_{j \in \text{NN of } i} \sum_{\sigma} \int_0^{\beta} d\tau \left(c_{i\sigma}^{\dagger}(\tau) c_{j\sigma}(\tau) + \text{h.c.} \right) + J \sum_{i \in \bar{0}} \sum_{j \in \text{NN of } i} \int_0^{\beta} d\tau \vec{S}_i(\tau) \cdot \vec{S}_j(\tau) \\ = S_{\text{int}}^{\text{hop}} + S_{\text{int}}^{\text{spin}} \quad (11.4)$$

In order to obtain an effective theory $S_{\text{eff}}^{0,\bar{0}}$ purely for the local system $(0, \bar{0})$, we need to trace over all the other degrees of freedom. This partial trace will be carried out over the states of the so-called ‘‘cavity’’ system $S^{(0,\bar{0})}$. The effective action can be constructed by tracing over scattering processes of all orders that leave the final action diagonal in the system $(0, \bar{0})$. The formal expression can be written as

$$S_{\text{eff}}^{0,\bar{0}} = S_{0,\bar{0}} + \sum_{n=1}^{\infty} \langle (S_{\text{int}}^{\text{hop}})^{2n} \rangle_{(0,\bar{0})} + \sum_{n=1}^{\infty} \langle (S_{\text{int}}^{\text{spin}})^{2n} \rangle_{(0,\bar{0})} \quad (11.5)$$

where, as mentioned before, the average is carried out in the cavity system $S^{(0,\bar{0})}$. Each hopping term in the expansion is of the form

$$\left(S_{\text{int}}^{\text{hop}} \right)^{2n} \\ = t^{2n} \sum_{\sigma} \sum_{(i_1, i'_1), \dots, (i_n, i'_n)} \sum_{(j_1, j'_1), \dots, (j_n, j'_n)} \int_0^{\beta} d\tau_1 \dots d\tau_n d\tau'_1 \dots d\tau'_{j_n} \\ c_{i_1\sigma}^{\dagger}(\tau_1) c_{i_2\sigma}^{\dagger}(\tau_2) \dots c_{i_n\sigma}^{\dagger}(\tau_n) c_{i'_1\sigma}(\tau'_1) c_{i'_2\sigma}(\tau'_2) \dots c_{i'_n\sigma}(\tau'_n) \left\langle c_{j_1\sigma}(\tau_1) c_{j_2\sigma}(\tau_2) \dots c_{j_n\sigma} c_{j'_1\sigma}^{\dagger}(\tau'_1) \dots c_{j'_n\sigma}^{\dagger}(\tau'_n) \right\rangle_{(0,\bar{0})} \\ = t^{2n} \sum_{\sigma} \sum_{(i_1, i'_1), \dots, (i_n, i'_n)} \sum_{(j_1, j'_1), \dots, (j_n, j'_n)} \int_0^{\beta} d\tau_1 \dots d\tau'_{j_n} c_{i_1\sigma}^{\dagger}(\tau_1) \dots c_{i_n\sigma}^{\dagger}(\tau_n) c_{i'_1\sigma}(\tau'_1) \dots c_{i'_n\sigma}(\tau'_n) \\ G_n^{(0,\bar{0})}(\tau_1 \dots \tau_n; \tau'_1 \dots \tau'_n), \quad (11.6)$$

where the indices i_1 through i_n and their primed counterparts run through $\bar{0}$, while $j_l (\in \{j_1, \dots, j'_n\})$ runs through the nearest-neighbours of the corresponding i_l index. The n -particle Greens function $G_n^{(0,\bar{0})}(\tau_1 \dots \tau_n; \tau'_1 \dots \tau'_n)$ for the cavity system is defined in the usual fashion

$$G_n^{(0,\bar{0})}(\tau_1 \dots \tau_n; \tau'_1 \dots \tau'_n) = \left\langle c_{j_1\sigma}(\tau_1) c_{j_2\sigma}(\tau_2) \dots c_{j_n\sigma} c_{j'_1\sigma}^\dagger(\tau'_1) \dots c_{j'_n\sigma}^\dagger(\tau'_n) \right\rangle_{(0,\bar{0})} \quad (11.7)$$

In the limit of infinite coordination number, only the $n = 1$ term survives:

$$\begin{aligned} \lim_{Z \rightarrow \infty} \sum_{n=1}^{\infty} \langle (S_{\text{int}}^{\text{hop}})^{2n} \rangle_{(0,\bar{0})} &= t^2 \sum_{\sigma} \sum_{i,i'} \int_0^{\beta} d\tau d\tau' c_{i\sigma}^\dagger(\tau) c_{i'\sigma}(\tau') \sum_{j,j'} G_1^{(0,\bar{0})}(j\tau; j'\tau') \\ &= \sum_{\sigma} \sum_{i,i'} \int_0^{\beta} d\tau d\tau' \Delta_{ii',\sigma}(\tau - \tau') c_{i\sigma}^\dagger(\tau) c_{i'\sigma}(\tau'), \end{aligned} \quad (11.8)$$

where we have defined the bath hybridisation function $\Delta_{ii',\sigma}(\tau - \tau')$ as

$$\Delta_{ii',\sigma}(\tau - \tau') = t^2 \sum_{j \in \text{NN of } i} \sum_{j' \in \text{NN of } i'} G_1^{(0,\bar{0})}(j\sigma\tau; j'\sigma\tau'). \quad (11.9)$$

Under similar approximations, the spin term in the action reduces to

$$\lim_{Z \rightarrow \infty} \sum_{n=1}^{\infty} \langle (S_{\text{int}}^{\text{spin}})^{2n} \rangle_{(0,\bar{0})} = \sum_{i,i'} \int_0^{\beta} d\tau d\tau' \chi(\tau - \tau') \vec{S}_i(\tau) \cdot \vec{S}_{i'}(\tau'), \quad (11.10)$$

where the susceptibility $\chi_{ii'}(\tau - \tau')$ is defined as

$$\chi_{ii'}(\tau - \tau') = J^2 \sum_{j \in \text{NN of } i} \sum_{j' \in \text{NN of } i'} \vec{S}_j(\tau) \cdot \vec{S}_{j'}(\tau'). \quad (11.11)$$

The effective action for the $0, \bar{0}$ system, in the limit of large coordination number, simplifies to

$$\begin{aligned} S_{\text{eff}}^{0,\bar{0}} &= \sum_{i=0,\bar{0}} \sum_{\sigma} \int_0^{\beta} d\tau c_{i\sigma}^\dagger(\tau) \partial_{\tau} c_{i\sigma}(\tau) - t \sum_{\bar{0},\sigma} \int_0^{\beta} d\tau \left[c_{0\sigma}^\dagger(\tau) c_{\bar{0}\sigma}(\tau) + \text{h.c.} \right] \\ &\quad - \frac{U}{2} \sum_{i=0,\bar{0}} \int_0^{\beta} d\tau (n_{i\uparrow}(\tau) - n_{i\downarrow}(\tau))^2 + J \sum_{\bar{0}} \int_0^{\beta} d\tau \vec{S}_0(\tau) \cdot \vec{S}_{\bar{0}}(\tau) \\ &\quad + \sum_{i,i'} \int_0^{\beta} d\tau d\tau' \left[\sum_{\sigma} \Delta_{ii',\sigma}(\tau - \tau') c_{i\sigma}^\dagger(\tau) c_{i'\sigma}(\tau') + \chi_{ii'}(\tau - \tau') \vec{S}_i(\tau) \cdot \vec{S}_{i'}(\tau') \right] \end{aligned} \quad (11.12)$$

One can also write down an effective action purely for the local site 0. The sites $\bar{0}$ will now be a part of the environment action $S^{(0)}$ instead of the system action, but since the system is thermodynamically large, the cavity action can be assumed to remain the same, such the expectation values are calculated in the same system. As a result, the hybridisation and susceptibility functions remain

unchanged. S_{int} is now made of terms that connect 0 and $\bar{0}$, so the transformation from $S^{(0,\bar{0})}$ can be made by replacing the set $\{j\}$ with the set $\bar{0}$, and the set $\bar{0}$ itself will be replaced by the local site 0. We quote the final form of the local effective action:

$$S_{\text{eff}}^0 = \sum_{\sigma} \int_0^{\beta} d\tau c_{0\sigma}^{\dagger}(\tau) \partial_{\tau} c_{0\sigma}(\tau) - \frac{U}{2} \int_0^{\beta} d\tau (n_{0\uparrow}(\tau) - n_{0\downarrow}(\tau))^2 + \int_0^{\beta} d\tau d\tau' \left[\sum_{\sigma} \Delta_{00,\sigma}(\tau - \tau') c_{i\sigma}^{\dagger}(\tau) c_{i'\sigma}(\tau') + \chi_{00}(\tau - \tau') \vec{S}_i(\tau) \cdot \vec{S}_{i'}(\tau') \right] \quad (11.13)$$

1.11.2 Local theory for the extended SIAM

The Hamiltonian for the extended SIAM model is shown in eq. ???. We will denote the impurity with the label d and the bath sites with the label c . Among the bath sites, we will represent the site coupled to the impurity with z . The full action has the form

$$S_{\text{ES}} = \sum_{d,c} \sum_{\sigma} \int_0^{\beta} d\tau c_{i\sigma}^{\dagger}(\tau) \partial_{\tau} c_{i\sigma}(\tau) - t \sum_{\langle i,j \rangle \in c, \sigma} \int_0^{\beta} d\tau \left[c_{i\sigma}^{\dagger}(\tau) c_{j\sigma}(\tau) + \text{h.c.} \right] - V \sum_{\sigma} \int_0^{\beta} d\tau \left[c_{d\sigma}^{\dagger}(\tau) c_{z\sigma}(\tau) + \text{h.c.} \right] + J \int_0^{\beta} d\tau \vec{S}_d(\tau) \cdot \vec{S}_z(\tau) - \frac{U}{2} \int_0^{\beta} d\tau (n_{d\uparrow}(\tau) - n_{d\downarrow}(\tau))^2 - \frac{W}{2} \int_0^{\beta} d\tau (n_{z\uparrow}(\tau) - n_{z\downarrow}(\tau))^2, \quad (11.14)$$

As in the Hubbard-Heisenberg model, we first obtain an effective action for the pair of sites (d, z) . As in the previous calculation, this will again generate a hybridisation function $\Delta_{zz,\sigma}$ for the bath site nearest to the zeroth site. No susceptibility will be generated however, because there is no spin-exchange coupling within the bath. The net result is

$$S_{\text{eff}}^{d,z} = \sum_{d,c} \sum_{\sigma} \int_0^{\beta} d\tau c_{i\sigma}^{\dagger}(\tau) \partial_{\tau} c_{i\sigma}(\tau) - V \sum_{\sigma} \int_0^{\beta} d\tau \left[c_{d\sigma}^{\dagger}(\tau) c_{z\sigma}(\tau) + \text{h.c.} \right] + J \int_0^{\beta} d\tau \vec{S}_d(\tau) \cdot \vec{S}_z(\tau) - \frac{U}{2} \int_0^{\beta} d\tau (n_{d\uparrow}(\tau) - n_{d\downarrow}(\tau))^2 - \frac{W}{2} \int_0^{\beta} d\tau (n_{z\uparrow}(\tau) - n_{z\downarrow}(\tau))^2 + \int_0^{\beta} d\tau d\tau' \sum_{\sigma} \Delta_{zz,\sigma}(\tau - \tau') c_{z\sigma}^{\dagger}(\tau) c_{z\sigma}(\tau') \quad (11.15)$$

This hybridisation $\Delta_{zz,\sigma}$ is calculated in the cavity model of the eSIAM, obtained by removing the impurity and the zeroth site from the full model; it is a completely non-interacting system.

One can go one step further and also remove the zeroth site in order to obtain a theory for the impurity site. With this choice, the cavity model now also includes the zeroth site, and hence contains the correlation term associated with W . The one-particle connection V will lead to a modified hybridisation $\mathcal{F}$ for this effective theory; the Kondo terms will generate a susceptibility. The resultant

action is

$$\begin{aligned}
S_{\text{eff}}^d = & \sum_{\sigma} \int_0^{\beta} d\tau \, c_{d\sigma}^{\dagger}(\tau) \partial_{\tau} c_{d\sigma}(\tau) - \frac{U}{2} \int_0^{\beta} d\tau \, (n_{d\uparrow}(\tau) - n_{d\downarrow}(\tau))^2 \\
& + \int_0^{\beta} d\tau d\tau' \left[\sum_{\sigma} \mathcal{F}_{\sigma}(\tau - \tau') c_{i\sigma}^{\dagger}(\tau) c_{i'\sigma}(\tau') + \chi_d(\tau - \tau') \vec{S}_i(\tau) \cdot \vec{S}_{i'}(\tau') \right]
\end{aligned} \tag{11.16}$$

where $\mathcal{F}$ is defined in terms of the local correlator of the zeroth site

$$\mathcal{F}_{\sigma}(\tau - \tau') = V^2 G_1^{(d)}(z_{\sigma}\tau; z_{\sigma}\tau') = V^2 \langle c_{z\sigma}(\tau) c_{z\sigma}^{\dagger}(\tau') \rangle_{(d)} \tag{11.17}$$

with the average computed in the interacting cavity model that has the impurity site removed. The interaction arises from the W -term. The susceptibility is also calculated in this interacting cavity model, and follows the same definition as in the bulk model:

$$\chi_d(\tau - \tau') = J^2 \vec{S}_z(\tau) \cdot \vec{S}_z(\tau') . \tag{11.18}$$